MIL-A-45899 (AR)
7 July 1977
SUPERSEDING
See Section 6

MILITARY SPECIFICATION

ADAPTION KIT, SUBASSEMBLY, ATOMIC WEAPON: XM7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of adaption kit subassembly entitled Adaption Kit, Subassembly, Atomic Weapon: XM7 which is used in the Sprint warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A- 50953	- Arm-Safe Device: XM88.
MIL-B- 48008	- Battery: XM97.
MIL-I- 50298	- Interconnecting Box: XM53.
MIL-S- 50563	- Support: XM9.
MIL-S- 48896	- Switching Unit, Safing: XM70.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

FSC 1135

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DRAWINGS

ORDNANCE CORPS

8888420	- Adaption Kit Subassembly, Atomic Weapon: XM7.
8888421	- Assembly Procedures for Adaption Kit Subassembly, Atomic Weapon: XM7.
9294760	- Shipping and Storage Container XM543E1, Packing and Marking, For Adaption Kit, Subassembly, Atomic Weapon: XM7.
EL-8888420	- Equipment List.

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1	- Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Adaption Kit (AK) subassembly shall comply with all requirements specified on the drawing (dwg) 8888420, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.2.1 Verification of major component status and assembly procedures. The AK subassembly shall be assembled in accordance with dwg 8888421. Verification of major components, assembly procedures, inspections and tests shall be in accordance with 3.7.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

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3.4 Serialization. A serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in Section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Major components. The contractor shall furnish objective evidence (see 6.2) that the following major components comply with the applicable specification:

Components	Specifications
Battery: XM97	MIL-B-48008
Support, Adaption Kit: XM9	MIL-S-50563
Interconnecting Box: XM53	MIL-I-50298
Arm-Safe Device: XM88	MIL-A-50953
Switching Unit, Safing: XM70	MIL-S-48896

3.7 Verification and test procedures. The following verification and test procedures are required during assembly of the AK subassembly to assure continuity of circuits and proper functioning and installation of components which are inaccessable after assembly.

3.7.1 Verification of major component status prior to start of assembly (see dwg 8888421 Note 1). The circuit resistance between points listed in Table I shall be as specified when tested in accordance with 4.4.1. Test current shall be not greater than 10 milliamperes. Resistance readings obtained shall be recorded in the inspection record.

Table I

Drawing Note	Component Part No.	From Pin	To Pin	Normal Condition Resistance Ohms
1A	9248944	13	17	0.100 max
* 1B	9274509	M1	M2	0.100 max
1C	9277000	M	N	0.100 max
1D	8888408	J2-A	J3-H	9.5K to 10.6K
		J2-B	J3-b	9.5K to 10.6K
		J2-J	J2-K	326 to 399

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*In addition to the circuit resistance test, specified herein, perform the following insulation test.

Note: Shorting strips must be between both positive terminals to both negative terminals prior to hypot. Short strips must also be applied from Scom to S_1 and S_2 respectively. The shorting strips are normally attached to the battery terminals by the manufacturer prior to shipping.

a. Positive (+) to Match (Scom)
b. Positive (+) to Monitor (M_1)
c. Positive (+) to Case
d. Monitor (M_1) to Match (Scom)
e. Monitor (M_1) to case
f. Match (Scom) to case

The insulation resistance between each isolated circuit and all others and case shall be 50 megohms minimum when tested with 500 ± 50 volts direct current (vdc) between the points specified in a. thru f. The insulation resistance shall be measured in accordance with MIL-STD-202, Method 302, Test Condition B for 10 seconds minimum or until the meter stabilizes. The current shall be limited to 5 milliamperes maximum. The battery voltage terminals shall be electrically connected together and treated as one electrically isolated circuit. The battery activation terminals shall be treated in the same manner.

3.7.2 Installation of connector, receptacle, electrical, Part No. 9243709 (see dwg 8888421, Note 4).

3.7.2.1 Torque. Using the equipment specified on the EL, verify the connector nut torque using 20 ± 1 in.-lbs of torque.

3.7.2.2 Circuit resistance. The circuit resistance between the connector, receptacle, electrical and a point on the XM9 Support, approximately 1 inch away from the connector, shall be not greater than 0.100 ohms, when tested as specified in 4.4.1.

3.7.3 Ground assembly torque (see dwg 8888421, Note 5). After assembling and installing grounding assembly verify socket head screw torque, 9 ± 1 in.-lbs using the equipment specified on the EL.

3.7.3.1 Circuit resistance (after installing ground assembly) (see dwg 8888421, Notes 5 and 6). The circuit resistance between connector, receptacle, electrical, Part No. 9243709 and the points indicated in Table II shall be as specified when tested in accordance with 4.4.1.

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Table II

Note No.	Circuit	Resistance (Ohms)
5 (Prior to Coating)	W1	0.100
	W2	0.100
6 (After Coating)	W1	0.100
	W2	0.100

3.7.4 Cap screw torque (After assembling BT1B battery assembly, arm-safe devices and IC box to XM9 support)(see dwg 8888421, Notes 7, 8 and 10B). After assembling the battery assembly, arm-safe device and IC box to the XM9 support verify cap screw torque 26 ± 1 in.-lbs, using the equipment specified on the EL.

3.7.5 Cap screw torque (After assembling BT1A battery assembly to XM9 support) (see dwg 8888421, Note 18). After assembling the battery assembly to the XM9 support verify cap screw torque 26 ± 1 in.-lbs, using the equipment specified on the EL.

3.7.6 Insulation resistance (After soldering flexible cable assembly No. 5 to BT1B header) (see dwg 8888421, Note 26). The insulation resistance between connector receptacle, electrical Part No. 9243709 and all other circuits as shown in Table III, shall be not less than 10 megohms when tested as specified in 4.4.2. Test voltage shall be 100 ± 5 vdc. Test current shall be not greater than 10 milliamperes.

Table III

Component	Pins
J2	G, R
Circuit Card Assembly A	E18, E21

3.7.7 Connector Nut torque (J1, J2, J3, J4) (see dwg 8888421 Note 29). Using the equipment specified on the EL, verify the connector nut torques as shown in Table IV.

Table IV

Connector	Torque
J1	20 ± 2 ft-lb
J2	25 ± 2 ft-lb
J3	30 ± 2 ft-lb
J4	10 + 1 - 0 ft-lb

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3.7.8 Cap screw torque (After installing XM70 and clamp, cable retaining, Part No. 9281737) (see dwg 8888421 Notes 31 and 32). After installing the XM70 and cable clamp verify cap screw torques, 26 + 1 in.-lbs for XM70 mounting screws and 7 + 1 in.-lbs for cable clamp screw, using the equipment specified on the EL.

3.7.9 Circuit resistance.

3.7.9.1 Prior to S10 and relay functional test (see dwg 8888421, Note 33). Circuit resistance shall be as specified in Table V when tested as specified in 4.4.1. Test current shall be not greater than 10 milliamperes.

Table V

From	To	Resistance (Ohms)
J1 Connector Shell	Connector P/N 9243709	0.100 max
J2 Connector Shell	Connector P/N 9243709	0.100 max
J3 Connector Shell	Connector P/N 9243709	0.100 max
J4 Connector Shell	Connector P/N 9243709	0.100 max
XM53-Terminal E143	Connector P/N 9243709	0.100 max
XM53-Terminal E229	Connector P/N 9243709	0.100 max
J2-A	J2-B	1.9K to 2.1K
J2-C	J2-Z	97.66K to 107.94K
J2-J	J2-K	11.8 to 14.8
J2-J	J2-c	10 Meg Min
J2-A	J2-$\overline{E}$	10 Meg Min
J2-$\overline{B}$	J2-F	10 Meg Min
J3-$\overline{H}$	J3-B	17.271K to 19.089K
XM53-Terminal E21	J1-$\overline{c}$	10 Meg Min
XM53-Terminal E21	J3-$\overline{D}$	10 Meg Min
XM53-Terminal E18	J1-B	10 Meg Min
XM53-Terminal E18	J3-$\overline{G}$	10 Meg Min

3.7.9.2 S10 and relays functional test. With voltage as shown in Figure 1 applied to J2-K(+) and J2-J(-) items S10, K1A and K1B shall operate. K1A and K1B operation shall cause circuit resistance within items 1 and 2 of Table VI, to be as shown. S10 operation shall cause a circuit resistance within items 3 thru 11 Table VI to be as specified when tested in accordance with 4.4.1. Circuit resistance test current shall be not greater than 10 milliamperes.

Table VI

Item	From	To	Resistance (Ohms)
1	J2-A	J2-E	1.8 to 2.1
2	J2-B	J2-F	1.8 to 2.1

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Table VI (Cont'd)

Item	From	To	Resistance (Ohms)
3	J2-A	J2-B	10 Meg Min
4	J2-C	J2-Z	62.13K to 68.67K
5	J2-J	J2-K	326.2 to 398.8
6	J2-J	J2-c	11.4 to 14.7
7	J3-H	J3-B	10 Meg Min
8	XM53 Terminal E21	J1-c	0.1 Max
9	XM53 Terminal E21	J3-D	13.3K to 14.7K
10	XM53 Terminal E18	J1-B	0.1 Max
11	XM53 Terminal E18	J3-G	13.3K to 14.7K

3.7.9.3 S10 reset test. With 24 ± 4 vdc applied to J2-J(+) and J2-c(-) for 100 ± 10 milliseconds S10 shall transfer to the set position. S10 transfer shall cause the circuit resistance to be as specified in Table VII when tested as specified in 4.4.1. Circuit resistance test current shall not be greater than 10 milliamperes.

Table VII

From	To	Resistance (Ohms)
J2-A	J2-B	1.9K to 2.1K
J2-C	J2-Z	97.66K to 107.94K
J2-J	J2-K	11.8 to 14.8
J2-J	J2-c	10 Meg Min
J3-H	J3-B	17.27K to 19.09K
XM53 Terminal E21	J1-c	10 Meg Min
XM53 Terminal E21	J3-D	10 Meg Min
XM53 Terminal E18	J1-B	10 Meg Min
XM53 Terminal E18	J3-G	10 Meg Min

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3.8 Electrical tests.

3.8.1 Circuit resistance. Circuit resistance between points specified in Table VIII shall be within ± 5% of the specified value (unless otherwise defined) when tested as specified in 4.4.1. Test voltage polarity must be observed, where specified. Test current shall be not greater than 10 milliamperes.

Table VIII

Item	From	To	Resistance Ohms	Test Code (See 3.5 and Table XXII)
1	J2-A	J3-H	10.09K	8101--
2	J2-B	J3-b	10.09K	8102--
3	J2-C	J2-Z	102.8K	8103--
4	J2-F	J3-EE	0.2 max	8104--
5	J2-G	Conn. Recept. PN 9243709	0.2 max	8105--
6	J2-H	Conn. Recept. PN 9243709	0.2 max	8106--
7	J3-C	Conn. Recept. PN 9243709	34K	8107--
8	J2-R	Conn. Recept. PN 9243709	0.2 max	8108--
9	J2-S	Conn. Recept. PN 9243709	0.2 max	8109--
10	J2-a	Conn. Recept. PN 9243709	40.2K	8110--
11	J2-b	Conn. Recept. PN 9243709	40.2K	8111--
12	J2-J	J2-K	11.8 to 14.8	8112--
13	J2-J	J3-GG	0.2 max	8113--
14	J2-K	J3-BB	14K	8114--

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Table VIII (Cont'd)

Item	From	To	Resistance Ohms	Test Code (See 3.5 and Table XXII)
15	J2-P	J3-A	1K	8115--
16	J2-U	J3-B	1K	8116--
17	J2-a	J3-DD	14K	8117--
18	J2-f	J3-c	1K	8118--
19	J2-h	J3-R	1K	8119--
20	J3-C	J3-n	28K	8120--
21	J3-D	J1-c	14K	8121--
22	J3-E	J1-P	16.5K	8122--
23	J3-F	J1-L	14K	8123--
24	J3-G	J1-b	14K	8124--
25	J3-M(+)	J1-P(-)	10K to 25K	8125--
26	J3-M(-)	J1-P(+)	100K min	8126--
27	J3-M	J3-P	20K	8127--
28	J3-N	J3-f	2K	8128--
29	J3-S(+)	J3-N(-)	15K to 25K	8129--
30	J3-S(-)	J3-N(+)	100K min	8130--
31	J3-T	J3-U	33K	8131--
32	J3-V(+)	J3-f(-)	15K to 25K	8132--
33	J3-V(-)	J3-f(+)	100K min	8133--
34	J3-W	J1-H	9.76K	8134--
35	J3-X	J3-Y	20K	8135--
36	J3-Y(+)	J1-R(-)	10K to 25K	8136--

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Table VIII (Cont'd)

Item	From	To	Resistance Ohms	Test Code (See 3.5 and Table XXII)
37	J3-Y(-)	J1-R(+)	100K min	8137--
38	J3-e	J4-A	14K	8138--
39	J3-g	J1-J	9.76K	8139--
40	J3-h	J1-R	16.5K	8140--
41	J3-k	J4-E	14K	8141--
42	J3-m	J1-M	14K	8142--
43	J3-p	J3-x	0.4 max	8143--
44	J3-FF	Conn. Recept. PN 9243709	34K	8144--
45	J1-E	J1-F	20	8145--
46	J1-B	J1-C	0.2 max	8146--
47	J1-B	J4-B	0.2 max	8147--
48	J1-B	J4-C	0.2 max	8148--
*49	J2-Y	J2-D	2K	8149--
*50	J2-A	J2-B	2K	8150--

*Items 49 and 50 must be tested after all items in this paragraph have been tested.

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3.8.2 Insulation resistance. Insulation resistance between each pin and all other pins shorted together as specified in Table IX shall be not less than 10 megohms when tested as specified in 4.4.2. Test voltage shall be 100 ± 5 volts dc with current limited to 10 milliamperes max.

Table IX

J2-A, C, D, E, F, G, J, N, P, T, U, V, c, e, f, g, h, i,

J3-D, F, G, P, W, X, Z, d, e, g, k, m, p, q, r, w, y, z, AA, K and a.

3.9 Temperature shock. The AK subassembly shall withstand a temperature shock of 110 + 5 degrees Fahrenheit (°F) to -40 + 10°F to 110 + 5°F without evidence of physical damage (see 6.5) when tested as specified in 4.4.3.

3.10 Transportation vibration. The AK subassembly shall withstand transportation random vibration as defined on Figures 23, 24 and 25 along the major axis Y, Z and X (Figure 2) for a test time of 2 hours ± 5 minutes per axis. Testing shall be as specified in 4.4.4.

3.11 Circuit resistance, switching unit safing (S10) and relay functional test requirements and procedures after temperature shock and transportation vibration tests.

3.11.1 Circuit resistance.

3.11.1.1 Prior to S10 and relay functional test. Circuit resistance shall be as specified in Table X when tested as specified in 4.4.1. Test current shall not be greater than 10 milliamperes.

Table X

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
1	J2-A	J2-B	1.9K to 2.1K	1101--
2	J2-C	J2-Z	97.66K to 107.94K	1102--
3	J2-J	J2-K	11.8 to 14.8	1103--
4	J2-J	J2-c	10 meg min	1104--
5	J2-A	J2-E	10 meg min	1105--
6	J2-B	J2-F	10 meg min	1106--
7	J3-H	J3-B	17.271K to 19.089K	1107--

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3.11.1.2 S10 and relays functional test. With voltage as shown in Figure 1 applied to J2-K(+) and J2-J(-), items S10, K1A and K1B shall operate. K1A and K1B operation shall cause circuit resistance within items 1 and 2 of Table XI to be as shown. S10 operation shall cause the circuit resistance within items 3 thru 7 Table XI to be as specified when tested in accordance with 4.4.1. Circuit resistance test current shall be not greater than 10 milliamperes.

Table XI

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
1	J2-A	J2-E	1.8 to 2.1	1108--
2	J2-$\overline{B}$	J2-F	1.8 to 2.1	1109--
3	J2-$\overline{A}$	J2-B	10 meg min	1110--
4	J2-C	J2-Z	62.13K to 68.67K	1111--
5	J2-J	J2-K	326.2 to 398.8	1112--
6	J2-J	J2-c	11.4 to 14.7	1113--
7	J3-H	J3-$\overline{B}$	10 meg min	1114--

CAUTION: Do not perform S10 reset test after transportation vibration test.

3.11.1.3 S10 reset test. With 24 + 4 volts dc applied to J2-J(+) and J2-c(-) for 100 + 10 milliseconds, S10 shall transfer to the set position. $\overline{S10}$ transfer shall cause the circuit resistance to be as specified in Table XII when tested as specified in 4.4.1. Circuit resistance test current shall not be greater than 10 milliamperes.

Table XII

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
1	J2-A	J2-B	1.9K to 2.1K	1115--
2	J2-C	J2-Z	97.66K to 107.94K	1116--
3	J2-J	J2-K	11.8 to 14.8	1117--
4	J2-J	J2-C	10 meg min	1118--
5	J2-A	J2-$\overline{E}$	10 meg min	1119--
6	J2-$\overline{B}$	J2-F	10 meg min	1120--
7	J3-$\overline{H}$	J3-B	17.27K to 19.09K	1121--

Note: Last digits of test code numbers are 01 after temperature shock and 02 for after transportation vibration.

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3.12 Random vibration. The AK subassembly shall withstand random vibration as defined in Figure 26 along each of the major axes Y, Z and X (Figure 2) for a test time of 20 ± 2 seconds per axis. Testing shall be as specified in 4.4.5.

3.13 Circuit resistance and insulation resistance test requirements after random vibration test.

3.13.1 Circuit resistance. Circuit resistance shall be as specified in Table XIII when tested as specified in 4.4.1. Test current shall be not greater than 10 milliamperes.

Table XIII

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
1	J2-A	J3-H	10.09K	2101--
2	J2-B	J3-b	10.09K	2102--
3	J2-C	J2-Z	65.4K	2103--
4	J2-F	J3-EE	0.2 max	2104--
5	J2-G	Conn. Recept. PN 9243709	0.2 max	2105--
6	J2-H	Conn. Recept. PN 9243709	0.200 max	2106--
7	J3-C	Conn. Recept. PN 9243709	34K	2107--
8	J2-R	Conn. Recept. PN 9243709	0.2 max	2108--
9	J2-S	Conn. Recept. PN 9243709	0.2 max	2109--
10	J2-J	J3-GG	0.2 max	2110--
11	J2-K	J3-BB	14K	2111--
12	J2-P	J3-A	1K	2112--
13	J2-U	J3-B	1K	2113--
14	J2-a	J3-DD	14K	2114--

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Table XIII (Cont'd)

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
15	J2-f	J3-c	1K	2115--
16	J2-h	J3-R	1K	2116--
17	J3-C	J3-n	28K	2117--
18	J3-D	J1-c	14K	2118--
19	J3-E	J1-P	16.5K	2119--
20	J3-F	J1-L	14K	2120--
21	J3-G	J1-b	14K	2121--
22	J3-M(+)	J1-P(-)	10K to 25K	2122--
23	J3-M(-)	J1-P(+)	100K min	2123--
24	J3-M	J3-P	20K	2124--
25	J3-N	J3-f	2K	2125--
26	J3-S(+)	J3-N(-)	15K to 25K	2126--
27	J3-S(-)	J3-N(+)	100K min	2127--
28	J3-T	J3-U	33K	2128--
29	J3-V(+)	J3-f(-)	15K to 25K	2129--
30	J3-V(-)	J3-f(+)	100K min	2130--
31	J3-W	J1-H	9.76K	2131--
32	J3-X	J3-Y	20K	2132--
33	J3-Y(+)	J1-R(-)	10K to 25K	2133--
34	J3-Y(-)	J1-R(+)	100K min	2134--
35	J3-e	J4-A	14K	2135--
36	J3-g	J1-J	9.76K	2136--

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Table XIII (Cont'd)

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
37	J3-h	J1-R	16.5K	2137--
38	J3-k	J4-E	14K	2138--
39	J3-m	J1-M	14K	2139--
40	J3-p	J3-x	0.4 max	2140--
41	J1-E	J1-F	20	2141--
42	J1-B	J1-C	0.2 max	2142--
43	J1-B	J4-B	0.2 max	2143--
44	J1-B	J4-C	0.2 max	2144--
*45	J2-Y	J2-D	2K	2145--
*46	J2-A	J2-B	10 meg min	2146--

*Items 45 and 46 must be tested after all items in this table have been tested.

3.13.2 Insulation resistance. The insulation resistance between each pin and all other pins shorted together as specified in Table XIV below shall be not less than 10 megohms when tested as specified in 4.4.2. Test voltage shall be 100 ± 5 vdc with current limited to 10 milliamperes max.

Table XIV

J2-A, C, D, G, J, N, P, T, U, V, e, f, g, h, i, E, and F.

J3-D, F, G, K, P, W, X, Z, a, d, e, g, k, m, p, q, r, w, y, z, and AA.

3.14 Acceleration. With acceleration as shown in Figure 3, applied to the +X axis of the adaption kit subassembly (Figure 2), the switches listed in Table XV shall operate as specified in 4.4.6. Test voltage shall be 28 vdc maximum with current limited to 10 milliamperes maximum. Test voltage polarity must be observed, where specified.

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Table XV

Switch	Test Points From	Test Points To	Pretest Conditions	Time of Operation (in sec) (Switch Closure)	Switch Remains Closed After End of Test	Switch Opens After End of Test
S9A-S4G	J3-y	J3-x	Open	*TA		X
S9B-S4G	J3-q	J3-p	Open	**TB		X
S9A-S5G	J3-w	J3-x	Open	TA + 0.040 max		X
S9B-S5G	J3-AA	J3-p	Open	TB + 0.040 max		X
Enable-A	J3-f(+)	J3-V(-)	Open	***TC + 0.663 ± 0.041	X	
Enable-B	J3-N(+)	J3-S(-)	Open	***TC + 0.663 ± 0.041	X	
Arm-A	J3-f(+)	J3-X(-)	Open	***TC + 1.290 ± 0.072	X	
Arm-B	J3-N(+)	J3-P(-)	Open	***TC + 1.290 ± 0.072	X	
S9A-Dud	J3-z	J3-x	Open	Switch closes and opens ten times between TA + 0.470 and TA + 2.000		X
S9B-Dud	J3-r	J3-p	Open	Switch closes and opens ten times between TB + 0.470 and TB + 2.000		X

*TA = Time of closure of S9A-S4G shall be used as reference for operation of S9A-S5G, Enable A, Arm A, S9A-Dud.

**TB = Time of closure of S9B-S4G shall be used as reference for operation of S9B-S5G, Enable B, Arm B, S9B-Dud.

***TC = Earliest time of TA or TB. Test current shall be limited to 0.010 amperes max.

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3.15 Circuit resistance and insulation resistance test requirements after acceleration test.

3.15.1 Circuit resistance. Circuit resistance between points specified in Table XVI shall be within ± 5% of the specified value (unless otherwise defined) when tested as specified in 4.4.1. Test voltage polarity must be observed where specified. Test current shall be not greater than 10 milliamperes.

Table XVI

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
1	J2-A	J3-H	10.09K	3101--
2	J2-B	J3-b	10.09K	3102--
3	J2-C	J2-Z	65.4K	3103--
4	J2-F	J3-EE	0.2 max	3104--
5	J2-G	Conn. Recept. PN 9243709	0.2 max	3105--
6	J2-H	Conn. Recept. PN 9243709	0.200 max	3106--
7	J3-C	Conn. Recept. PN 9243709	34K	3107--
8	J2-R	Conn. Recept. PN 9243709	0.2 max	3108--
9	J2-S	Conn. Recept. PN 9243709	0.2 max	3109--
10	J2-J	J3-GG	0.2 max	3110--
11	J2-K	J3-BB	14K	3111--
12	J2-P	J3-A	1K	3112--
13	J2-U	J3-B	1K	3113--
14	J2-a	J3-DD	14K	3114--
15	J2-f	J3-c	1K	3115--

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Table XVI (Cont'd)

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
16	J2-h	J3-R	1K	3116--
17	J3-C	J3-n	28K	3117--
18	J3-D	J1-c	14K	3118--
19	J3-E	J1-P	16.5K	3119--
20	J3-F	J1-L	14K	3120--
21	J3-G	J1-b	14K	3121--
22	J3-M(+)	J1-P(-)	10K to 25K	3122--
23	J3-M(-)	J1-P(+)	100K min	3123--
24	J3-M	J3-P	20K	3124--
25	J3-N	J3-f	2K	3125--
26	J3-S(+)	J3-N(-)	15K	3126--
27	J3-S(-)	J3-N(+)	15K	3127--
28	J3-T	J3-U	33K	3128--
29	J3-V(+)	J3-f(-)	15K	3129--
30	J3-V(-)	J3-f(+)	15K	3130--
31	J3-W	J1-H	9.76K	3131--
32	J3-X	J3-Y	20K	3132--
33	J3-Y(+)	J1-R(-)	10K to 25K	3133--
34	J3-Y(-)	J1-R(+)	100K min	3134--
35	J3-e	J4-A	14K	3135--
36	J3-g	J1-J	9.76K	3136--
37	J3-h	J1-R	16.5K	3137--

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Table XVI (Cont'd)

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
38	J3-k	J4-E	14K	3138--
39	J3-m	J1-M	14K	3139--
40	J3-p	J3-x	0.4 max	3140--
41	J1-E	J1-F	20	3141--
42	J1-B	J1-C	0.2 max	3142--
43	J1-B	J4-B	0.2 max	3143--
44	J1-B	J4-C	0.2 max	3144--
45	J2-Y	J2-D	10 meg min	3145--
46	J2-A	J2-B	10 meg min	3146--

3.15.2 Circuit resistance. Upon completion of the circuit resistance tests per 3.15.1, perform operations 2 through 8 of drawing 8888411 and perform the circuit resistance tests specified in Table XVII below. Upon completion of the above perform operations specified in Notes 9 through 12 of drawing 8888411.

Table XVII

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
1	J2-a	J2-E	1.8 to 2.1	3201--
2	J2-b	J2-F	1.8 to 2.1	3202--
3	S9A-4	S9A-20	0.020 max	3203--
4	S9A-7	S9A-18	0.020 max	3204--
5	S9A-8	S9A-19	0.020 max	3205--
6	S9A-12	S9A-22	0.020 max	3206--

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Table XVII (Cont'd)

Item	From	To	Resistance (Ohms)	Test Code (See 3.5 and Table XXII)
7	S9B-4	S9B-20	0.020 max	3207--
8	S9B-7	S9B-18	0.020 max	3208--
9	S9B-8	S9B-19	0.020 max	3209--
10	S9B-12	S9B-22	0.020 max	3210--

3.15.3 Insulation resistance. Upon completion of 3.15.2, replace AK cover per drawing 8888411. Thereafter the insulation resistance between each pin and all other pins shorted together as specified in Table XVIII shall be not less than 10 megohms when tested as specified in 4.4.2. Test voltage shall be 100 ± 5 vdc with current limited to 10 milliamperes max.

Table XVIII

J2-A, C, D, G, J, N, P, T, U, V, e, f, g, h, and i.

J3-D, F, G, W, Z, d, e, g, k, m, p, q, r, w, y, z, AA, K and a.

3.16 Operation. After subjecting the AK subassembly to the requirements of 3.14, the AK subassembly shall be operated by applying the input signals as specified in Table XIX. With the above input signals applied to the AK subassembly, the events and output signals specified in Tables XX and XXI shall be obtained. Operate times and resistance readings shall be recorded. Testing shall be as specified in 4.4.6.1.

Caution: Adaption kit battery negative and positive are isolated above missile frame ground by static bleeder resistors. This isolation integrity is a system interface requirement and shall be maintained by system test equipment.

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Table XIX

Input Signals

Event	Time (In Seconds) Input Signal Application	Time (In Seconds) Input Signal Removal	Input Signal Voltage Applied From	Input Signal Voltage Applied To	Input Voltage Required	Instrumentation Points From	Instrumentation Points To
Input Signal #2A See Fig. 11	T_o + 0.01 max	T_2 + 0.055±0.005	J2-a(+)	J2-E&F(-)	*7 to 20 ** Volts dc	J3-DD	J3-EE
Input Signal #2B	T_o + 0.01 max	T_2 + 0.055±0.005	J2-b(+)	J2-E&F(-)	*7 to 20 Volts dc	--	--
Input Signal #3A	T_o + 58±1.01	T_3 + 0.025±0.005	J2-i	J2-V	Square Wave Voltage 25±5 Volts Peak-to-Peak	--	--
					Frequency 50±15KHz	--	--
Input Signal #3B See Fig. 12	T_o + 58±1.01	T_3 + 0.025±0.005	J2-U	J2-h	Square Wave*** Voltage 25±5 Volts Peak-to-Peak	J3-B	J3-R
					Frequency 50±15KHz	--	--
Input Signal #4A	T_3 + 0.405±0.005	T_4+0.0016±0.0006	J2-g	J2-T	Square Wave Voltage 25±5 Volts Peak-to-Peak	--	--
					Frequency 50±15KHz	--	--

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Table XIX Continued

Input Signals

Event	Time (In Seconds) Input Signal Application	Time (In Seconds) Input Signal Removal	Input Signal Voltage Applied From	Input Signal Voltage Applied To	Input Voltage Required	Instrumentation Points From	Instrumentation Points To
Input Signal #4B See Fig. 13	T_3 + 0.405±0.005	T_4+0.0016±0.0006	J2-P	J2-f	Square Wave*** Voltage 25±5 Volts Peak-to-Peak	J3-A	J3-c
					Frequency 50±15KHz	--	--
Input Signal #5	T_3 + 0.600±0.010	T_5 + 0.025±0.005	J2-e	J2-N	Square Wave Voltage 13±2.5 Volts Peak-to-Peak	--	--
					Frequency 50±15KHz	--	--

T_o = Start of test (This time is arbitrary. No electrical events occur at this time).

T_2 = Time of application of Input Signal #2A and #2B.

T_3 = Time of application of Input Signal #3A and #3B.

T_4 = Time of application of Input Signal #4A and #4B.

T_5 = Time of application of Input Signal #5.

EOT = End of Test = 62 + 2 seconds.

*May go to 32 vdc when battery matches and squib bridge wire burn open.

**An instrumentation frequency response of dc to 5KHz min is required.

***An instrumentation frequency response of dc to 500KHz min is required.

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Table XX

AK Subassembly Events

Event	Time (In Seconds) Event Voltage Applied To Instrumentation Point	Event Voltage Removed From Instrumentation Point	Instrumentation Point From	Instrumentation Point To	Resistance Value Required (Ohms)	Voltage Required At Instrumentation Point
Monitor Loop Enabled See Fig. 14	Recording of exact time of event not required. Visual indication of event occurrence is required.	Once enabled, Monitor loop remains enabled until after EOT	J2-C	J2-Z	2K ± 5% (65.4K±5% prior to event)	***
Battery A Voltage See Fig. 15	Approximately T_o + 0.2 (No specific voltage requirement until T_o + 0.7)	EOT	J3-V	J1-C	--	(After T_o + 0.7*** sec) **32.5±4.5 Volts dc
Battery B Voltage See Fig. 15	Approximately T_o + 0.2 (No specific voltage requirement until T_o + 0.7)	EOT	J3-S	J1-B	--	(After T_o + 0.7*** sec) **32.5±4.5 Volts dc
S7A Operate See Fig. 16	T_3 + 1.55±0.250	EOT	J3-Z	J1-C	--	32 +4 -9 Volts dc***
S7B Operate See Fig. 16	T_3 + 1.55±0.250	EOT	J3-d	J1-B	--	32 +4 -9 Volts dc***

*May go to 32 vdc when battery matches and squib bridge wire burn open.

**After application of output signals (#5A and #5B, Table XXI) voltage may decrease by 5.0 volts.

***An instrumentation frequency response dc to 5KHz min is required.

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Table XXI

Output Signals

Event	Application of Output Signal to Load (In Seconds)	Removal of Output Signal From Load (In Seconds)	Output Signal Voltage Applied From	Output Signal Voltage Applied To	Load Applied To Output Signal Pins	Output Signal Voltage Required At J3 & At Load (Volts dc)	Instrumentation Points (In Addition To Those Shown In Load Figures) From	Instrumentation Points (In Addition To Those Shown In Load Figures) To
Output Signal #1A See Fig.17	Approx. T_o+0.2 (No specific voltage requirement until T_o+0.7)	EOT	J4-A	J4-B	See Fig. 4	(After T_o+ 0.7 sec)* **32.5±4.5 ***	J3-e	J1-C
Output Signal #1B See Fig.17	Approx. T_o+0.2 (No specific voltage required until T_o+0.7)	EOT	J4-E	J4-C	See Fig. 4	(After T_o+ 0.7 sec)* **32.5±4.5 ***	J3-k	J1-B
Output Signal #2A See Fig.18	Approx. T_o+0.2 (No specific voltage required until T_o+0.7)	T_3+1.55±0.250	J1-R	J1-C	See Fig. 5	(After T_o+ 0.7 sec) *32.5±4.5 ***	J3-h	J1-C
Output Signal #2B See Fig.18	Approx. T_o+0.2 (No specific voltage required until T_o+0.7)	T_3+1.55±0.250	J1-P	J1-B	See Fig. 5	(After T_o+ 0.7 sec) *32.5±4.5 ***	J3-E	J1-B
Output Signal #3A See Fig.19	T_3+0.0012±0.0008	T_3+1.55±0.250	J1-b	J1-C	See Fig. 5	Must be in range shown in Fig. 10***	J3-G	J1-C
Output Signal #3B See Fig.19	T_3+0.0012±0.0008	T_3+1.55±0.250	J1-c	J1-B	See Fig. 6	Must be in range shown in Fig. 10***	J3-D	J1-B

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Table XXI Continued

Output Signals

Event	Application of Output Signal to Load (In Seconds)	Removal of Output Signal From Load (In Seconds)	Output Signal Voltage Applied From	Output Signal Voltage Applied To	Load Applied To Output Signal Pins	Output Signal Voltage Required At J3 & At Load (Volts dc)	Instrumentation Points (In Addition To Those Shown In Load Figures) From	Instrumentation Points (In Addition To Those Shown In Load Figures) To
Output Signal #4A See Fig.20	T_4+0.0012±0.0008	T_3+1.55±0.250	J1-J	J1-C	See Fig. 7	*31 ± 3***	J3-g	J1-C
Output Signal #4B See Fig.20	T_4+0.0012±0.0008	T_3+1.55±0.250	J1-H	J1-B	See Fig. 7	*31 ± 3***	J3-W	J1-B
Output Signal #5A See Fig.21	T_3+0.600±0.010	EOT	J1-F	J1-C	See Fig. 8	32$^{+4}_{-9}$ ***	J3-n	J1-C
Output Signal #5B See Fig.21	T_3+0.600±0.010	EOT	J1-E	J1-B	See Fig. 8	32$^{+4}_{-9}$ ***	J3-C	J1-B
Output Signal #6A See Fig.22	T_3+1.02±0.052	EOT	J1-M	J1-C	See Fig. 9	32$^{+4}_{-9}$ ***	J3-m	J1-C
Output Signal #6B See Fig.22	T_3+1.02±0.052	EOT	J1-L	J1-B	See Fig. 9	32$^{+4}_{-9}$ ***	J3-F	J1-B

*After application of output signals #5A and #5B voltage may decrease by 5.0 volts.

**May vary between 7 and 10 volts for the first 50 milliseconds after signal application.

***An instrumentation frequency response dc to 5KHz min is required.

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Table XXII

Test Condition	Last Digits
Ambient	00
After Temperature Shock	01
Trans Vibration	02
After Random Vibration	03
Acceleration	04

3.17 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.18 Workmanship.

3.18.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.18.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.18.3 Burrs. Parts shall be free from burrs.

3.18.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.18.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.18.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 1 complete assembly (see dwg 8888421) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table XXIII. Tests shall be conducted in order specified in Table XXIII.

Table XXIII

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of defects/characteristics (see 4.3.2.1)		X
2. Electrical tests* (see 3.8)		X
3. Temperature shock (see 3.9)	X	
4. Electrical tests* (see 3.8 and 3.11)		X
5. Transportation vibration (see 3.10)	X	
6. Electrical tests* (see 3.8 and 3.11 except 3.11.1.2)		X
7. Random vibration (see 3.12)	X	
8. Electrical tests* (see 3.13)		X
9. Acceleration (see 3.14)	X	
10. Electrical tests* (see 3.15)		X
11. Operation (see 3.16)		X

*Perform insulation resistance test before circuit resistance test.

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting

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safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor defects/characteristics table (see Table XXIV and Table XXV), and inspected for all defects/characteristics in the applicable classification. Any defects/characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects/characteristics in each classification need not be followed as long as all defects/characteristics inspections are performed.

Table XXIV
Major Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table XXV
Minor Defects/Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Adaption kit subassembly, XM7 (see dwg 8888420).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Components missing or damaged	Visual
102.	Any component assembled incorrectly or loose	Visual
103.	Screw improperly torqued (32 places)	Gage
104.	Inner gaskets protrudes more than 0.025" beyond cover	Gage
105.	Cover not sealed properly	Visual
106.	Objective evidence that gasket material is as specified	Visual
107.	Lockwires and seals missing or incorrectly installed	Visual
108.	Evidence of poor workmanship (see 3.18)	Visual
109.	Identification plate missing, damaged, mislocated or illegible	Visual
110.	Objective evidence that all major components conform to their respective specifications or drawings, missing (see 3.1)	Visual
111.	Outer gasket protrudes more than 1/16" beyond cover	Gage
112.	Any hole in inner gasket misaligned with through holes in XM9 Support (13 places)	Visual
*113.	Existence of cracks at through holes with inserts (inner and outer flanges)	Visual
114.	Serialization improper or missing (see 3.4)	Visual
Minor:	None defined.	

*Use a minimum of 5X magnification for this inspection.

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4.3.2.1.2 Preparation for delivery (see 5).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Preservation, packing, and packaging not in accordance with dwg 9294760	Visual
102.	Marking not in accordance with dwg 9294760	Visual
Minor:		
201.	Evidence of poor workmanship (see 3.18)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	Insulation resistance	Major
b.	Circuit resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance and insulation resistance. Each AK subassembly in the lot shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.8.1 and 3.8.2, the item shall be rejected and removed from the lot.

4.3.3.2 Sampling. One*sample shall be selected at random during the second months production. Thereafter one sample shall be selected, at random, from every 35 units or portion thereof produced. The samples selected shall be tested in accordance with the requirements of Table XXIII items 3 through 10. If the sample fails to comply with the applicable requirements of Table XXIII items 3 through 10, the contractor will suspend shipment of any further production units until such time as the following condition has been met:

The contractor analyses the cause of failure, takes corrective action, using procedures approved by the Government, retests the corrected unit, or a new production unit, to the requirements of Table XXIII items 3 through 10 and the unit passes all the required testing of Table XXIII items 3 through 10.

*Monthly sample shall consist of all new components except XM88E1, A/S Devices and, XM9 Support.

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4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8888420). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. The circuit resistance shall be measured in accordance with 3.7.1, 3.8.1, 3.11.1, 3.13.1 or 3.15.1 as applicable, with approved measuring equipment.

4.4.2 Insulation resistance. The insulation resistance shall be measured with approved insulation resistance measuring equipment in accordance with 3.7.6, 3.8.2, 3.13.2 or 3.15.3, as applicable.

4.4.3 Temperature shock. The AK subassembly shall be placed in a test chamber that has been stabilized (see 6.4) at 110 ± 5°F, and maintained for a minimum of 4 hours. The AK subassembly shall then be removed from the high temperature chamber and within 3 minutes placed in a chamber that has been stabilized (see 6.4) at -40 ± 5°F and maintained at this temperature for a minimum of 4 hours. Within 3 minutes, the AK subassembly shall be returned to the high temperature chamber. This sequence constitutes one cycle. Three consecutive cycles shall be executed. Upon completion of the third cycle, hold the test unit in the test chamber stabilized at 110 ± 5°F for a minimum of 4 hours, then return the unit to ambient room temperature (80 ± 5°F) without forced cooling, and test in accordance with 3.8.1, 3.8.2 and 3.11 in this order.

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4.4.4 Transportation vibration. The AK subassembly shall be mounted to a fixture capable of being oriented readily in each of three mutually perpendicular axes. The fixture and the AK subassembly shall be vibrated over the frequency ranges specified in Figures 23, 24 and 25. After this test the AK subassembly shall be tested as specified in 3.8.1, 3.8.2 and 3.11 in this order.

4.4.5 Random vibration. The AK subassembly shall be mounted to a fixture capable of being oriented readily in each of the three mutually perpendicular axes. The fixture and the AK subassembly shall be vibrated over the frequency range specified in Figure 26. After this test, the AK subassembly shall be tested as specified in 3.13.

4.4.6 Acceleration. The AK subassembly shall be placed in an approved shockless fixture and secured to an approved centrifuge. With the centrifuge at the 90 ± 5g level, the major input axis X (see Figure 2) must be swung from minus to plus within 100 milliseconds. During this test, monitor AK subassembly for the requirements specified in Table XIX. Upon completion of the above, remove the AK subassembly from the centrifuge and test as specified in 3.15.

4.4.6.1 Operation. After completion of tests specified in 3.15, function and monitor AK as shown in Tables XX and XXI.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The AK subassembly shall be packaged in accordance with dwg 9257235 using the specified moisture barrier.

5.1.2 Level C. The AK subassembly shall be packaged in accordance with dwg 9257234 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The AK subassembly shall be packed in accordance with dwg 9257235.

5.3 Marking. Marking shall be as specified in accordance with dwg 9257235.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

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6.1.2 Contract data requirements (see 3.5 and 4.3.4.1). Data specified in 3.5 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Physical damage. Physical damage shall include bent or loose contact pins, split or cracked insulation, damaged seal interfaces, or cracks in the body.

6.6 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2188 dated 20 June 1973 and PA-PD-2188A dated 15 July 1974, and SCN 1 thru 3.

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Project No. 1135-A220

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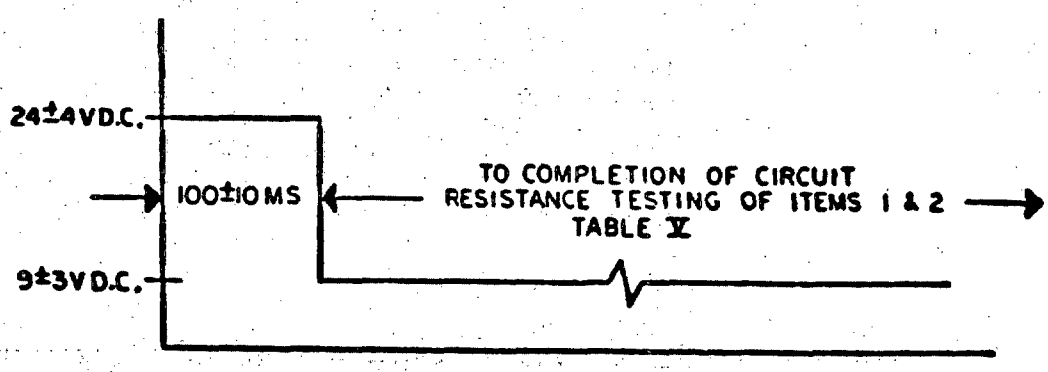


Figure 1. S10 and Relays Functional Test Voltage Requirements

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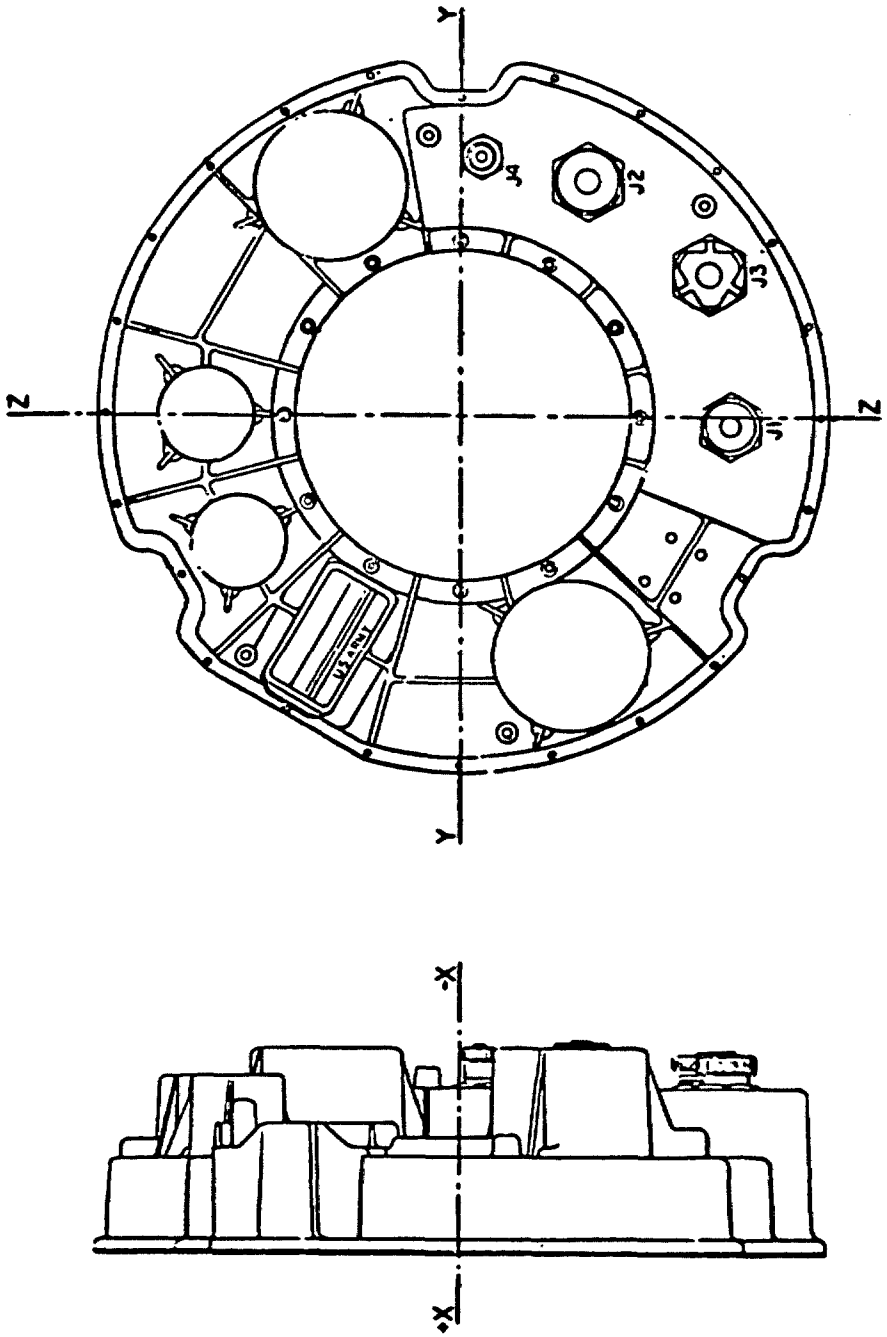


Figure 2. XM7 AK Subassembly Axis Determination

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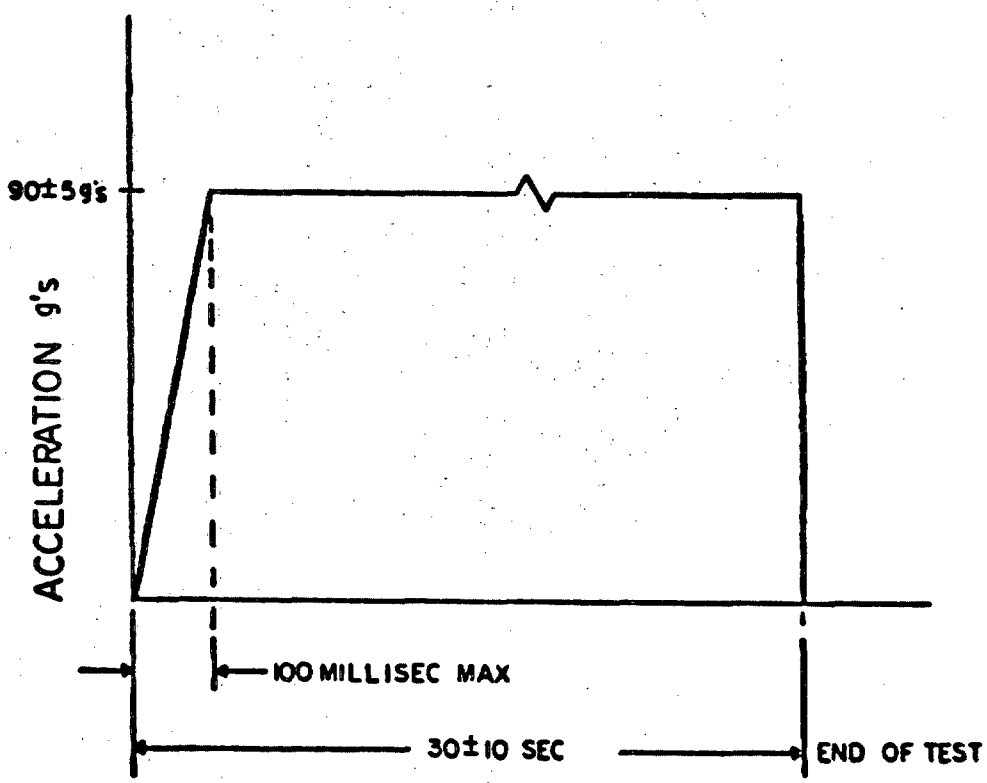
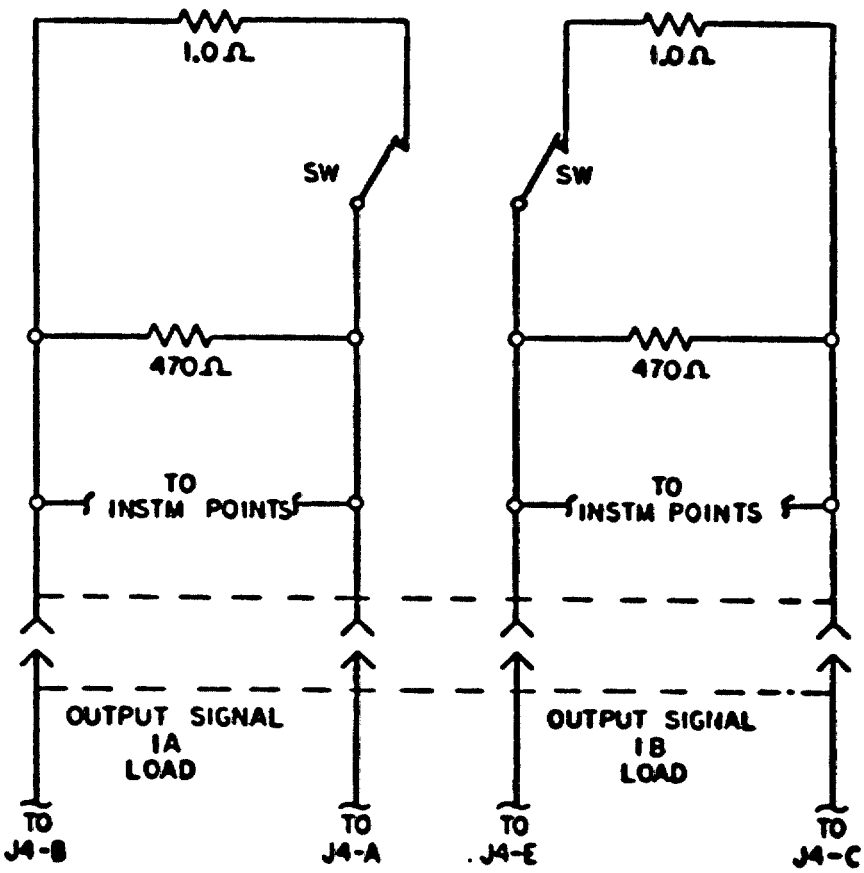


Figure 3. Acceleration Time Profile

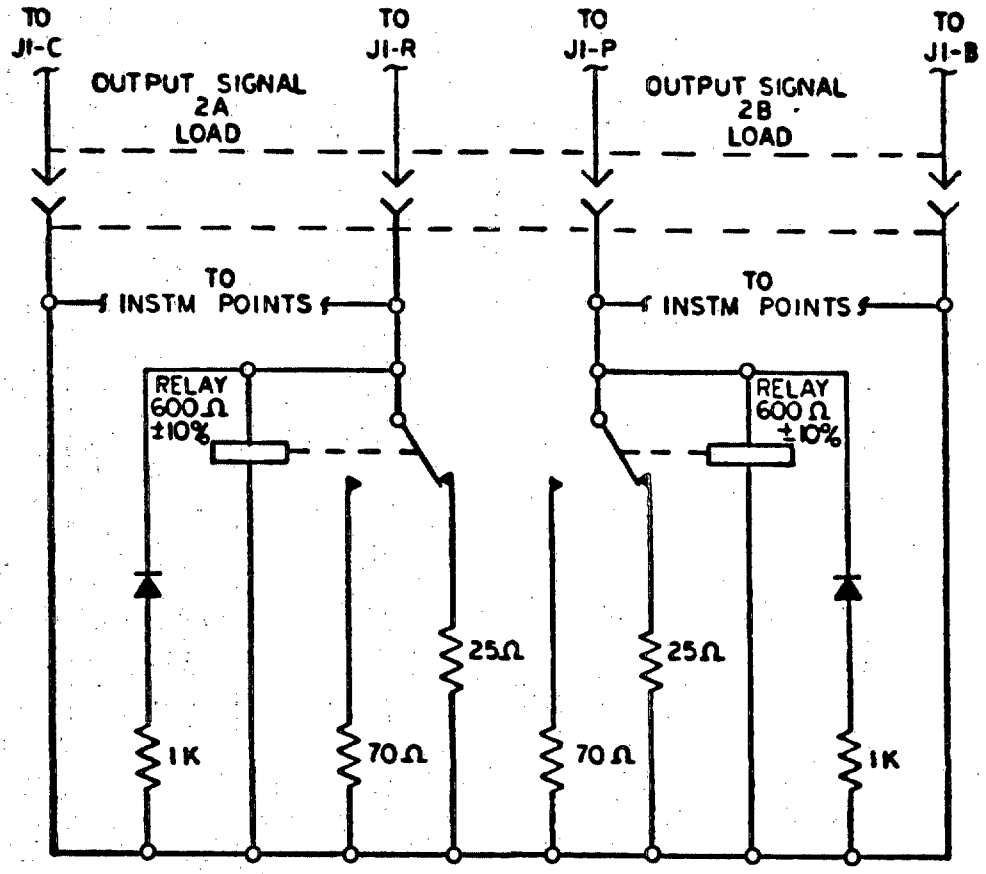
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NOTES:
1. SWITCH SHALL BE NORMALLY CLOSED.
2. SWITCH CONTACT RESISTANCE SHALL BE 0.050 Ω MAX.
3. SWITCH SHALL OPERATE TO THE OPEN POS IN 55 + 5 MILLISECONDS AFTER THE OUTPUT SIGNAL IS APPLIED TO THE LOAD AND SHALL REMAIN IN THE OPEN POS UNTIL END OF TEST.
4. TOLERANCE ON RESISTORS ± 5%.

Figure 4. Equivalent Circuit Output Signal 1A & 1B Loads

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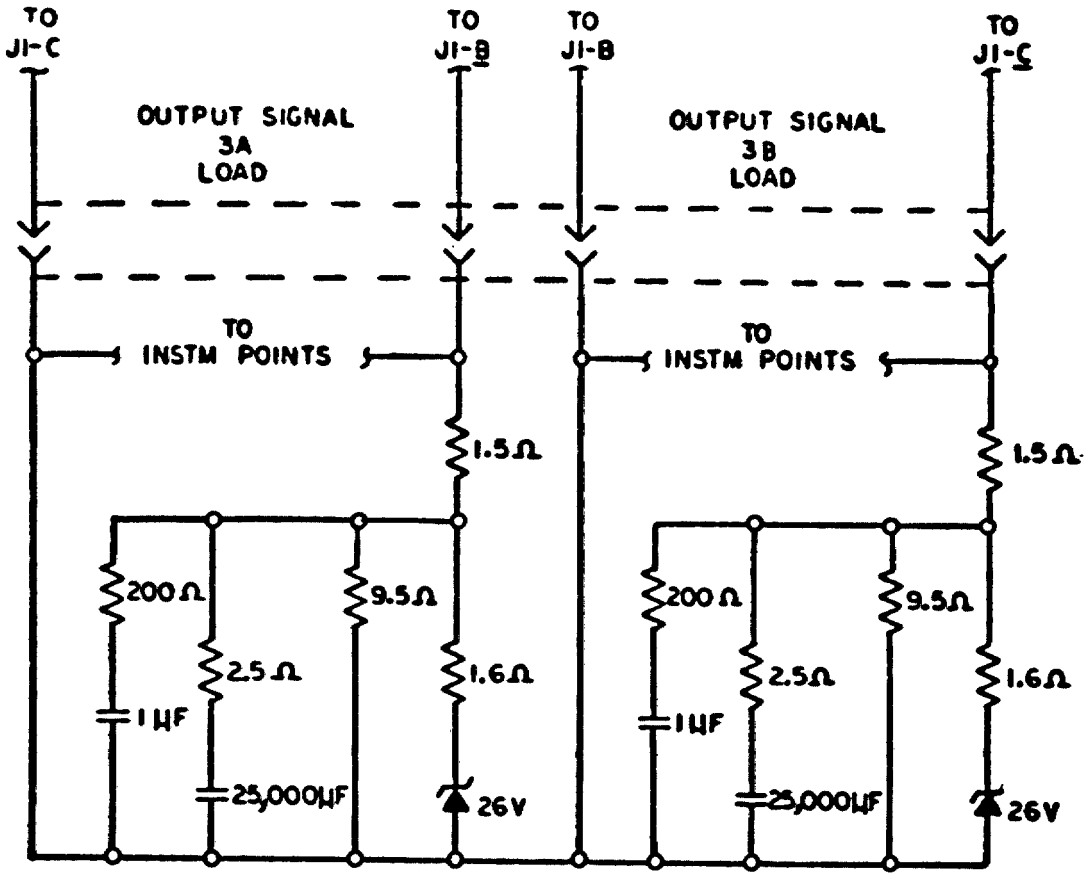


NOTES:

1. RELAY SHALL OPERATE IN 4 + 2 MILLISECONDS AFTER APPLICATION OF OUTPUT SIGNAL TO LOAD AND SHALL REMAIN OPERATED UNTIL END OF TEST.
2. RELAY CONTACT RESISTANCE SHALL BE 0.050 Ω MAX.
3. TOLERANCE ON RESISTORS ± 5%.

Figure 5. Equivalent Circuit Output Signal 2A & 2B Loads

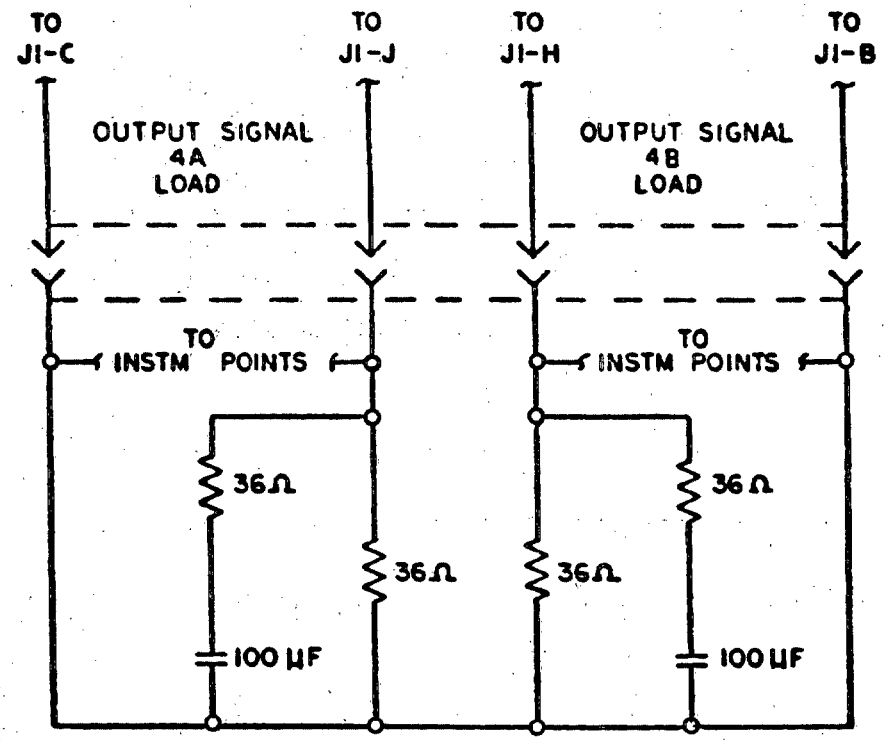
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NOTES:
1. TOLERANCE: ON CAPACITORS +25%-10%, ON RESISTORS ± 5%.

Figure 6. Equivalent Circuit Output Signal 3A & 3B Loads

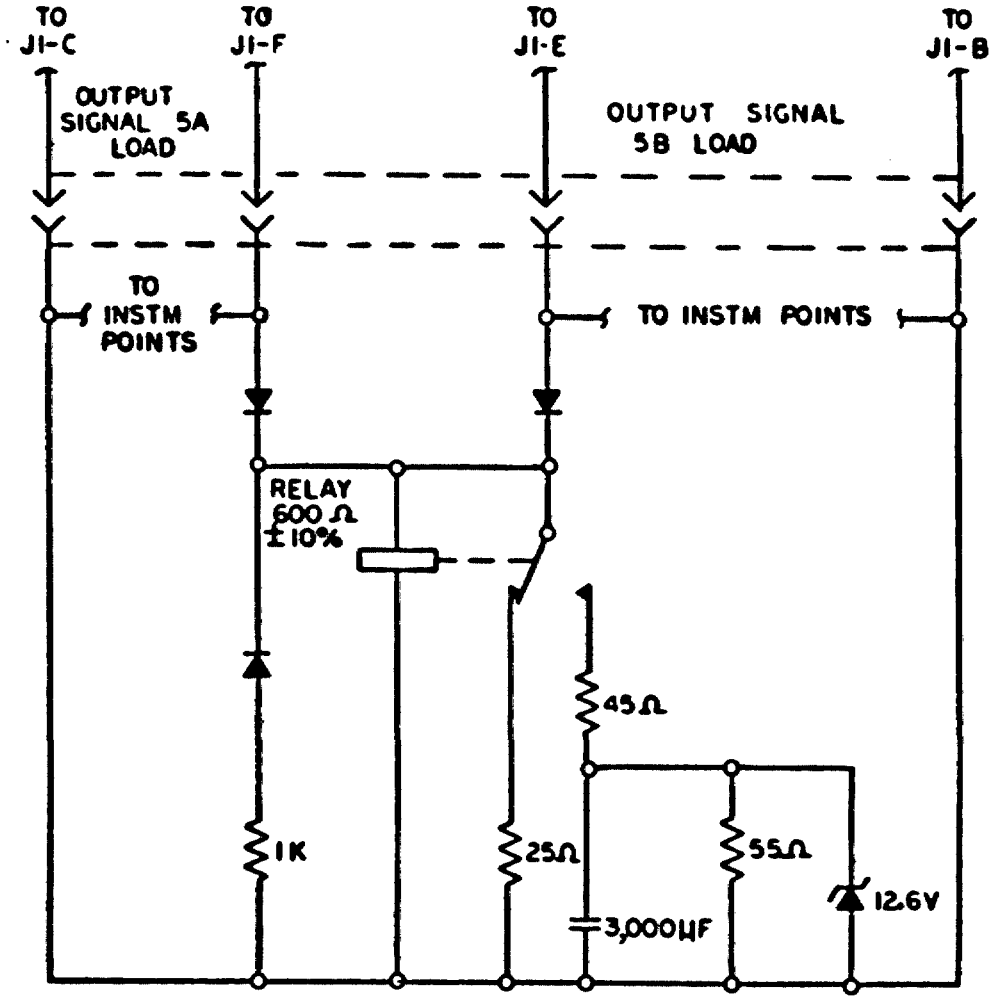
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:07Z
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NOTES:
1. TOLERANCE: ON CAPACITORS +25%-10%, ON RESISTORS ± 5%.

Figure 7. Equivalent Circuit Output Signal 4A & 4B Loads

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NOTES:

1. RELAY SHALL OPERATE IN 4 + 2 MILLISECONDS AFTER APPLICATION OF OUTPUT SIGNAL TO LOAD AND SHALL REMAIN OPERATED UNTIL END OF TEST.
2. RELAY CONTACT RESISTANCE SHALL BE 0.050 Ω MAX.
3. TOLERANCE ON CAPACITORS +25%-10% ON RESISTORS ± 5%.

Figure 8. Equivalent Circuit Output Signal 5A & 5B Loads

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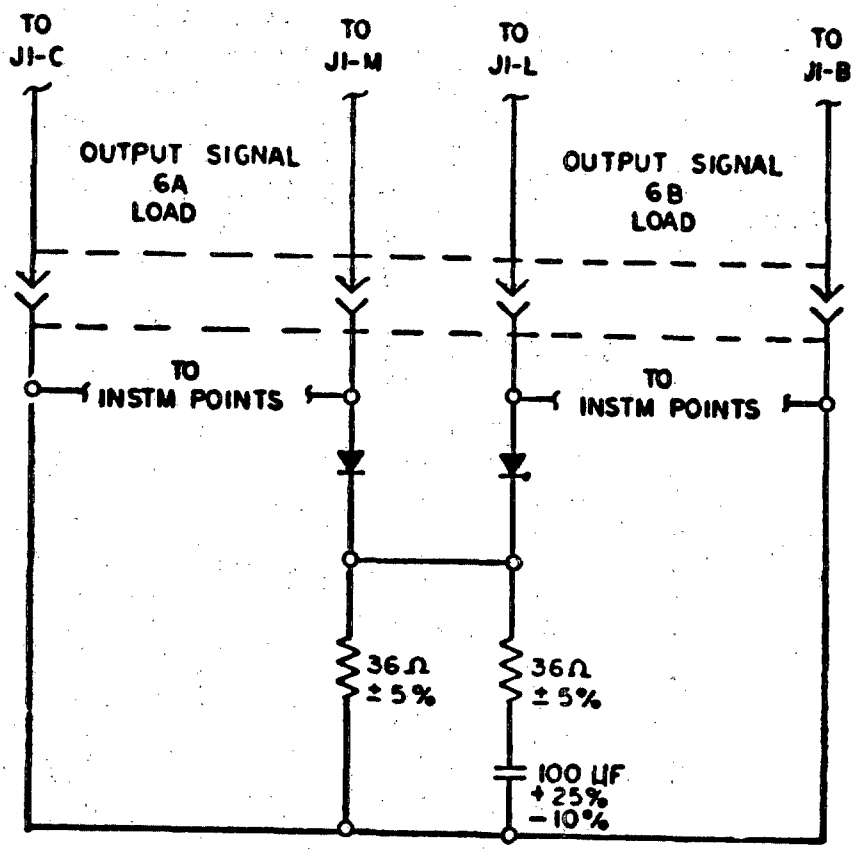


Figure 9. Equivalent Circuit Output Signal 6A & 6B Loads

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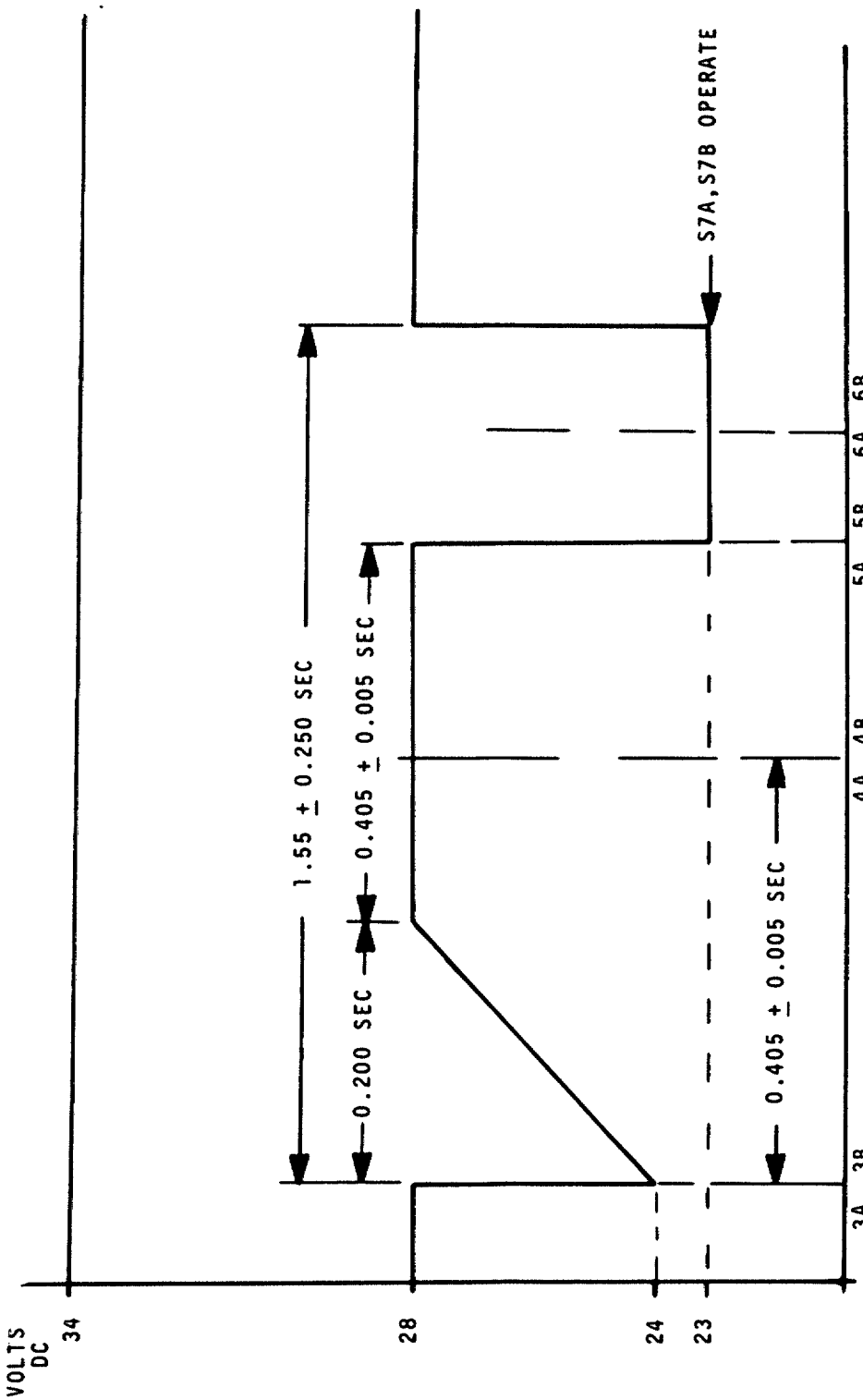
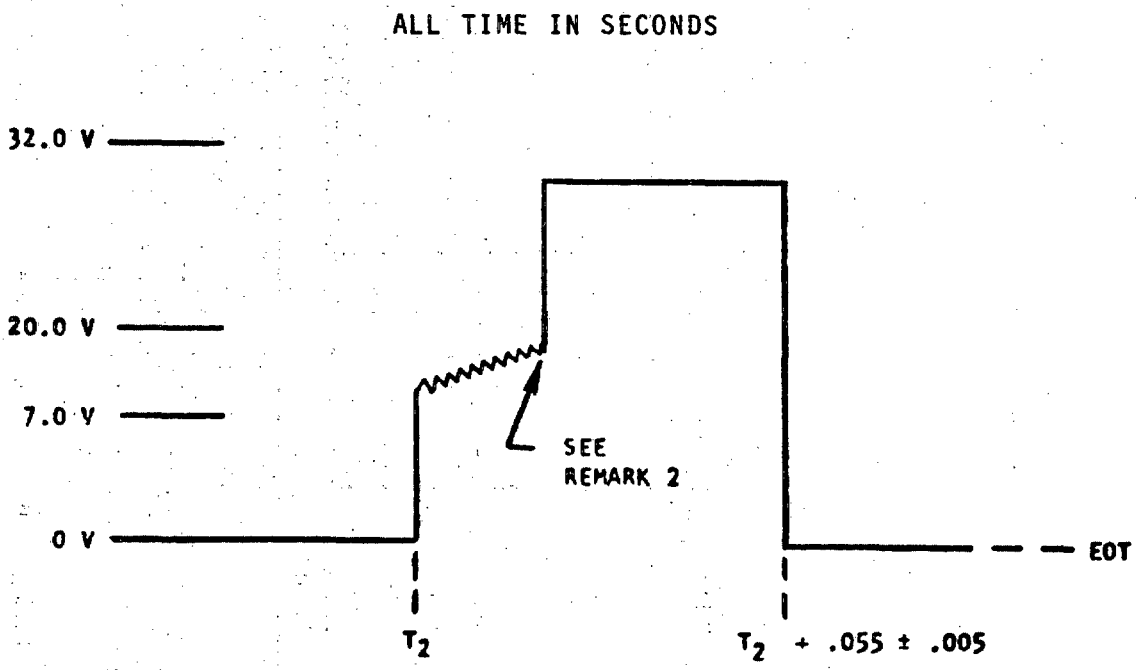


Figure 10. Voltage Profile Output 3A & 3B Signals

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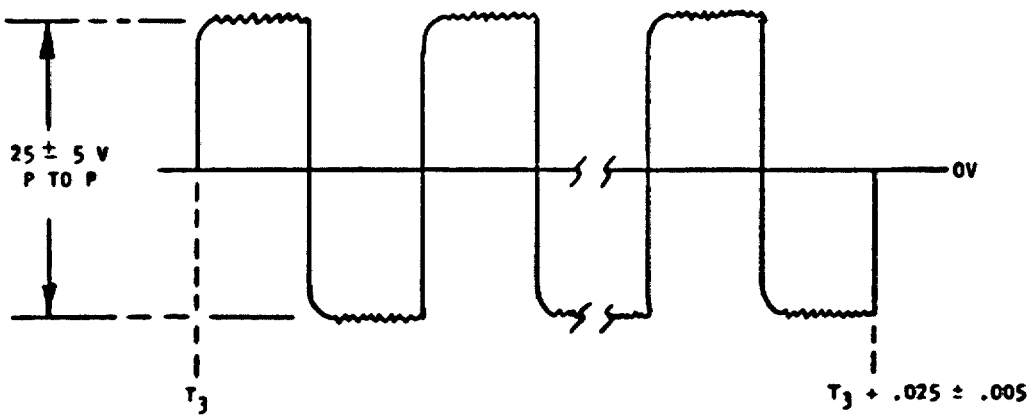
REMARKS: 1. $T_2 = T_0 + 0.01$ MAX.

2. STEP INDICATES BRIDGE WIRES HAVE BURNED OPEN.

Figure 11. Voltage Input 2A

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ALL TIME IN SECONDS



REMARKS: 1. $T_3 = T_0 + 58 \pm 1.01$.

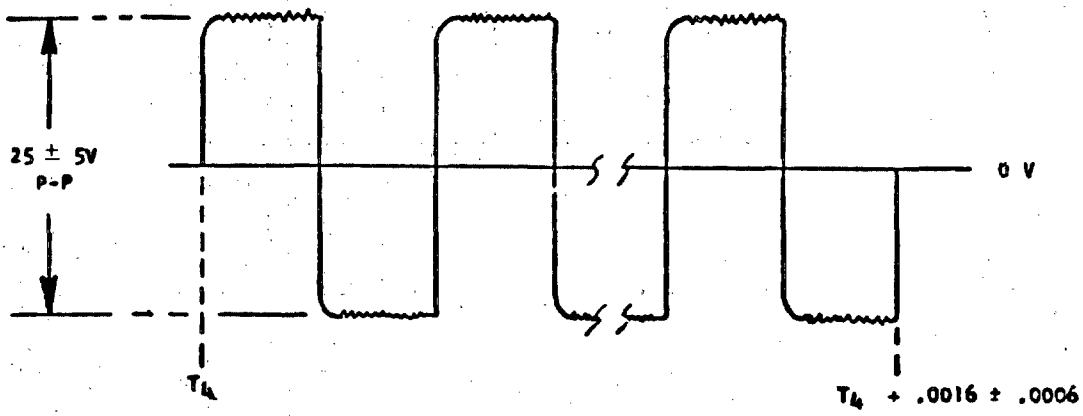
2. 50 KHZ + 15 KHZ SQUARE WAVE. SQUARE WAVE MAY START EITHER POSITIVE OR NEGATIVE. WAVE SHAPE FOR THE FIRST SEVERAL MILLISECONDS IS IMPORTANT SINCE SQUIB FUNCTIONS DURING THIS TIME.

3. SIGNAL SHALL BE ZERO AT ALL OTHER TIMES.

Figure 12. Voltage Input 3B

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ALL TIME IN SECONDS



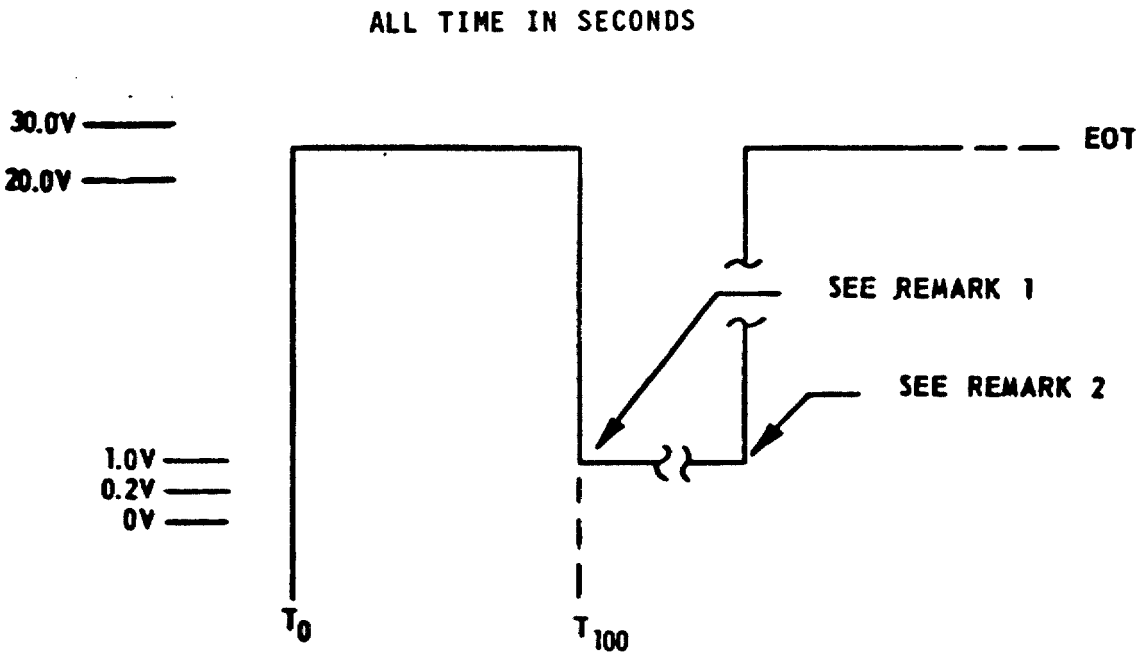
REMARKS: 1. $T_4 = T_3 + 0.405 \pm 0.005$.

2. 50 KHZ + 15 KHZ SQUARE WAVE. SQUARE WAVE MAY START EITHER POSITIVE OR NEGATIVE. WAVE SHAPE FOR THE FIRST SEVERAL MILLISECONDS IS IMPORTANT SINCE SQUIB FUNCTIONS DURING THIS TIME.

3. SIGNAL SHALL BE ZERO AT ALL OTHER TIMES.

Figure 13. Voltage Input 4B

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REMARKS:

1. VOLTAGE OF 0.2 TO 1.0 VOLTS INDICATES A CIRCUIT RESISTANCE OF 2000 OHMS.

2. VOLTAGE REMOVED BY SIMULATOR AT APPROXIMATELY T_0 + 110 MS.

3. A CHANGE IN RESISTANCE IN THE MONITOR LOOP ENABLE CIRCUIT IN THE AK IS REPRESENTED BY A CHANGE IN VOLTAGE IN THE SIMULATOR. THIS SIGNAL IS MONITORED AT THE SIMULATOR.

4. T_{100}

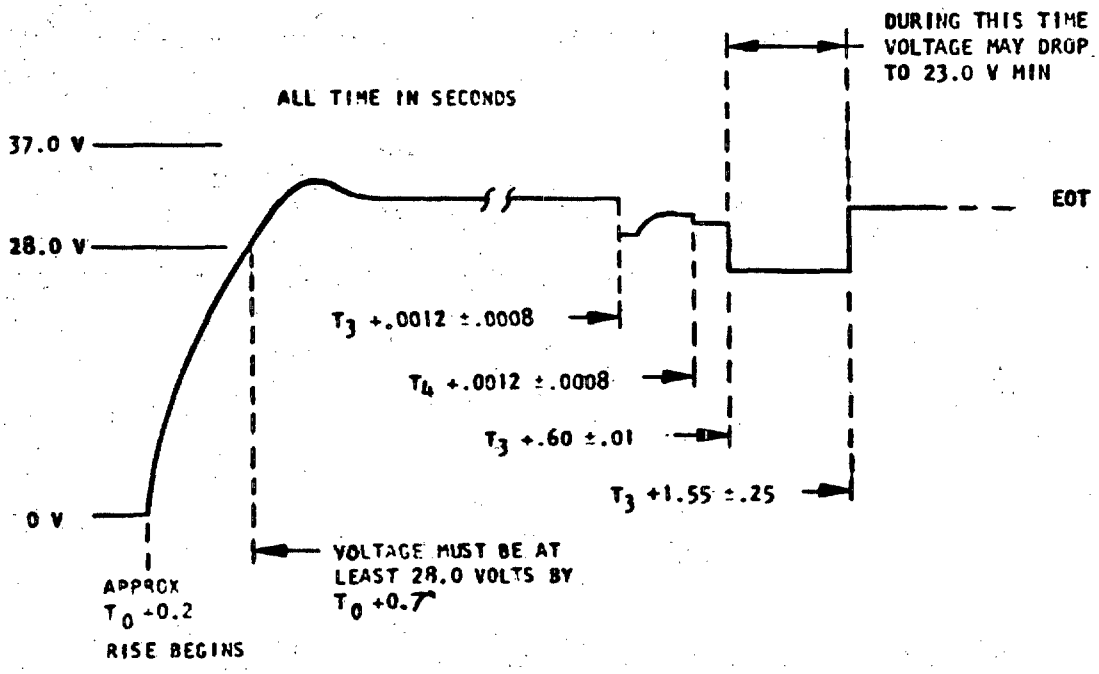
   SITUATION #1 REMOTE ARM SIGNAL GIVEN NORMALLY BY LPE SIMULATOR. $T_{100} = T_0$ + 100 MS MAX.

   SITUATION #2 REMOTE ARM SIGNAL INHIBITED BY PROVIDING N.O. SWITCH IN REMOTE ARMLINE IN LPE SIMULATOR. $T_{100} = T_0$ + 3 MS.

Figure 14. Voltage Monitor Loop Enable

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ALL TIME IN SECONDS



REMARKS:

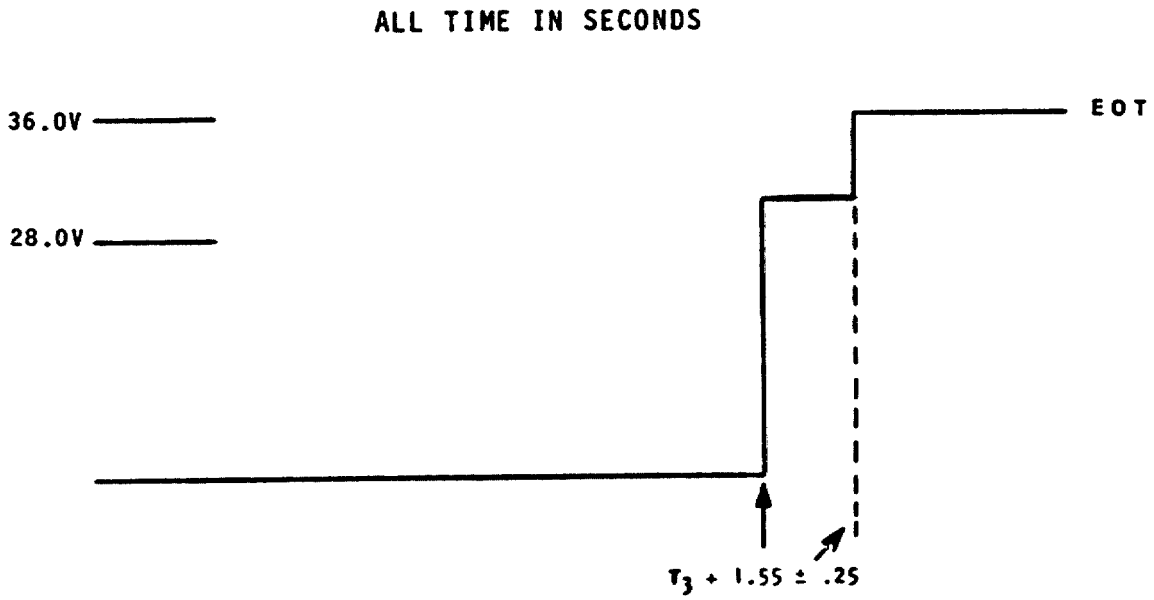
1. BATTERY VOLTAGE MUST BE BETWEEN 28.0 V AND 37.0 V FROM T_0 + 0.7 UNTIL T_0 + 60.0 EXCEPT FROM T_3 + 0.60 + 0.01 TO T_3 + 1.55 + 0.25 WHEN AN ADDITIONAL 5 VOLT DROP MAY OCCUR. THE ONLY OTHER SIGNIFICANT VARIATION OF VOLTAGE MAY OCCUR AT APPROXIMATELY T_0 + 2.0 WHEN BATTERY SWITCHES FROM FAST STACK TO SLOW STACK. THIS WILL RESULT IN A ONE OR TWO VOLT SHIFT BUT MUST REMAIN WITHIN THE 27.0 V TO 32.0 VOLT RANGE.

2. PRIOR TO T_0 + 700 MS, INSTRUMENTATION MAY NOT INDICATE ACTUAL BATTERY VOLTAGE DUE TO AN INCOMPLETE RETURN CIRCUIT.

3. BETWEEN T_3 + 0.0012 + 0.0008 AND T_4 + 0.0012 + 0.0008, THE SIGNAL SHALL BE WITHIN BOUNDS SPECIFIED FOR SIGNAL OUTPUT 3A AND 3B.

Figure 15. Voltage Battery "A" or Battery "B"

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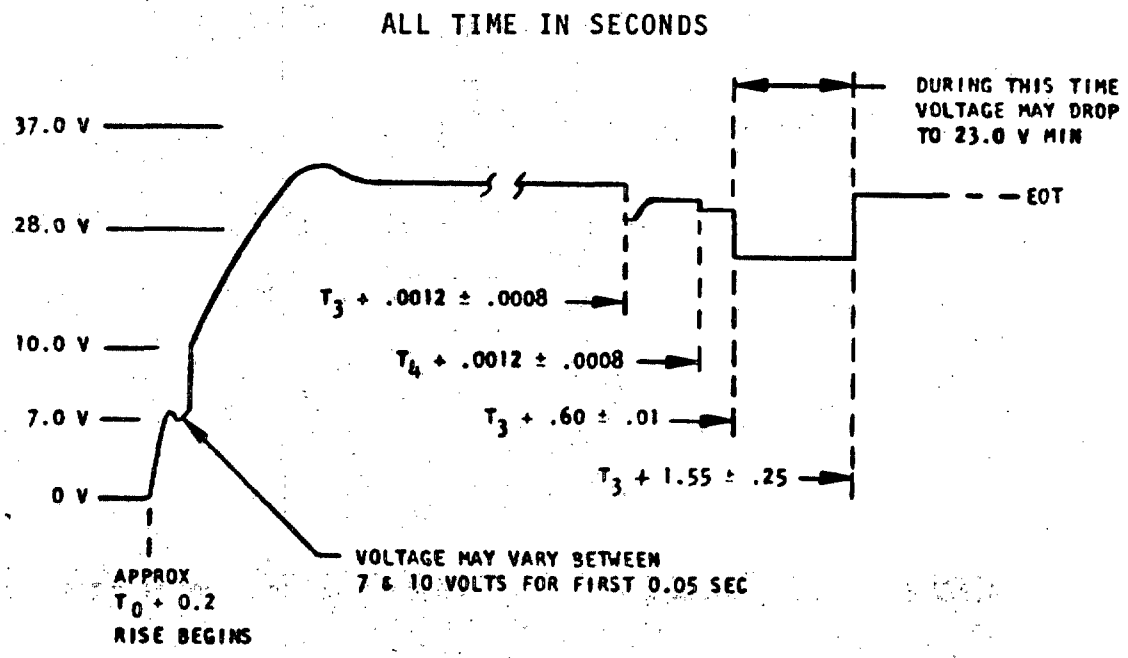
REMARKS: SIGNAL SHALL BE ZERO AT ALL OTHER TIMES.

1. FIRST STEP IS DUE TO ONE S7 SQUIB FUNCTION. FIRST STEP MAY BE HARD TO DISTINGUISH IF SQUIBS FUNCTION SIMULTANEOUSLY.

2. SECOND STEP IS DUE TO REMOVAL OF AK OUTPUT LOADS 2A, 2B, 3A, 3B, AND 4A AND 4B, WHEN SECOND S7 SQUIB FUNCTIONS.

Figure 16. Voltage S7A or S7B

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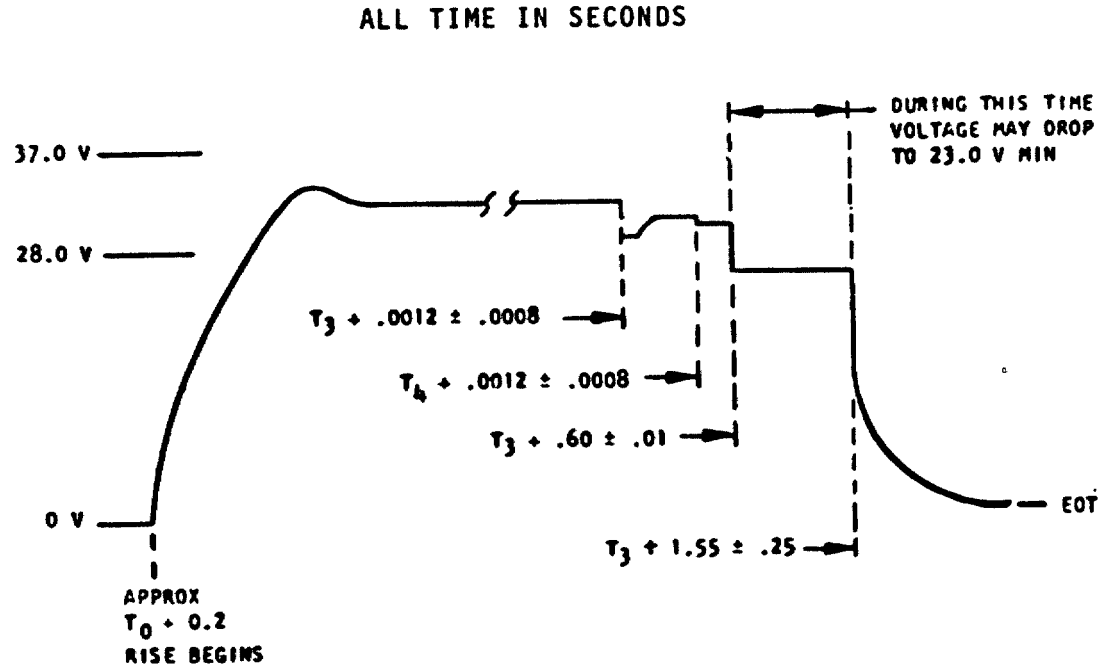


REMARKS:

BETWEEN T_3 + 0.0012 ± 0.0008 AND T_4 + 0.0012 ± 0.0008 THE SIGNAL SHALL BE WITHIN BOUNDS SPECIFIED FOR SIGNAL OUTPUT 3A AND 3B.

Figure 17. Voltage Output 1A and 1B (AK) or Output 1A and 1B (WH)

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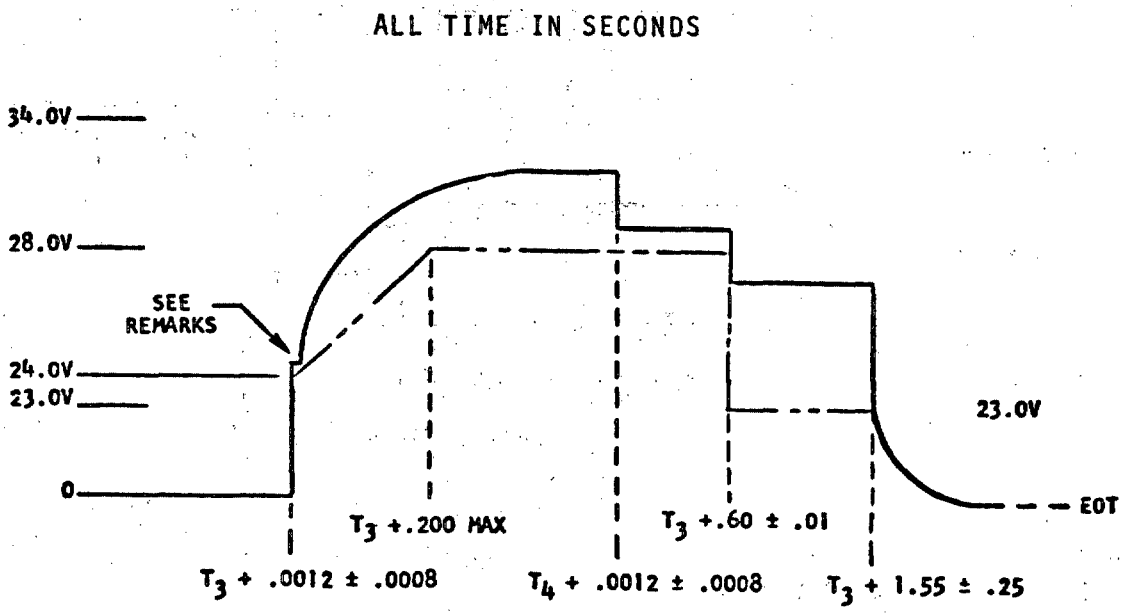


REMARKS:

1. SIGNAL SHALL BE ZERO AT ALL OTHER TIMES.
2. BETWEEN T_3 + 0.0012 $\pm$ 0.0008 AND T_4 + 0.0012 $\pm$ 0.0008 THE SIGNAL SHALL BE WITHIN BOUNDS SPECIFIED FOR SIGNAL OUTPUT 3A AND 3B.

Figure 18. Voltage Output 2A and 2B (AK) or Output 2A and 2B (WH)

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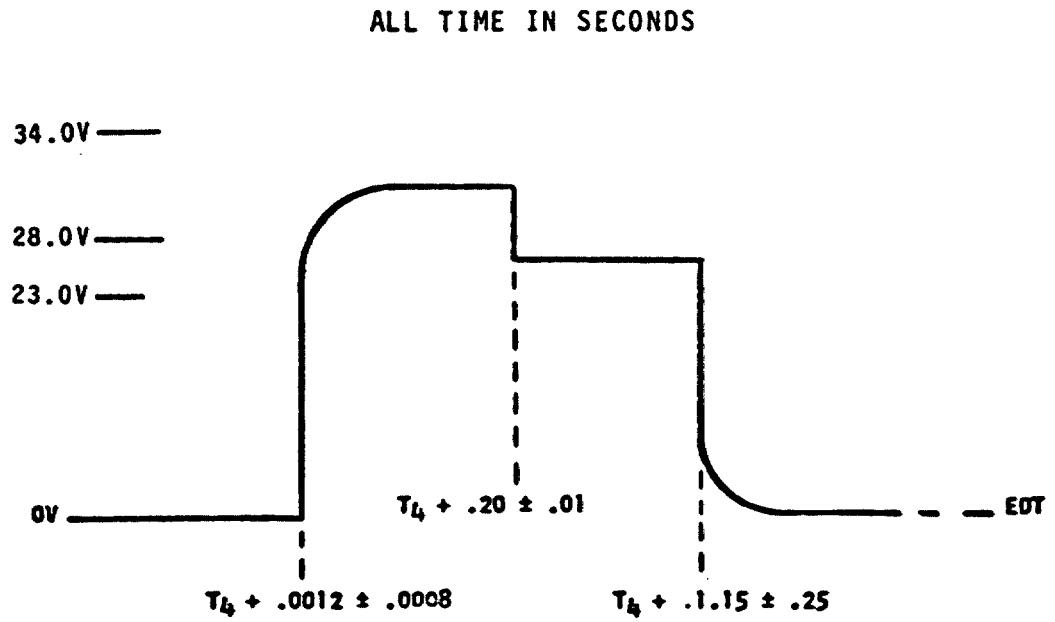
REMARKS: $T_4 = T_3 + 0.405 \pm 0.01$

1. A STEP SHOULD OCCUR ON THE LEADING EDGE OF 3A/3B SIGNAL. THIS LOAD IS PRESENT FOR SEVERAL MILLISECONDS AND WILL CHANGE WHEN THE S7 BRIDGE WIRES BURN OPEN. THE SIGNAL MUST BE ABOVE THE MINIMUM LEVELS AT ALL TIMES, EVEN WHEN THE S7 BRIDGE WIRE LOAD IS PRESENT.

2. THE SIGNAL SHALL BE ZERO AT ALL OTHER TIMES.

Figure 19. Voltage Output 3A and 3B (AK) or Output 3A and 3B (WH)

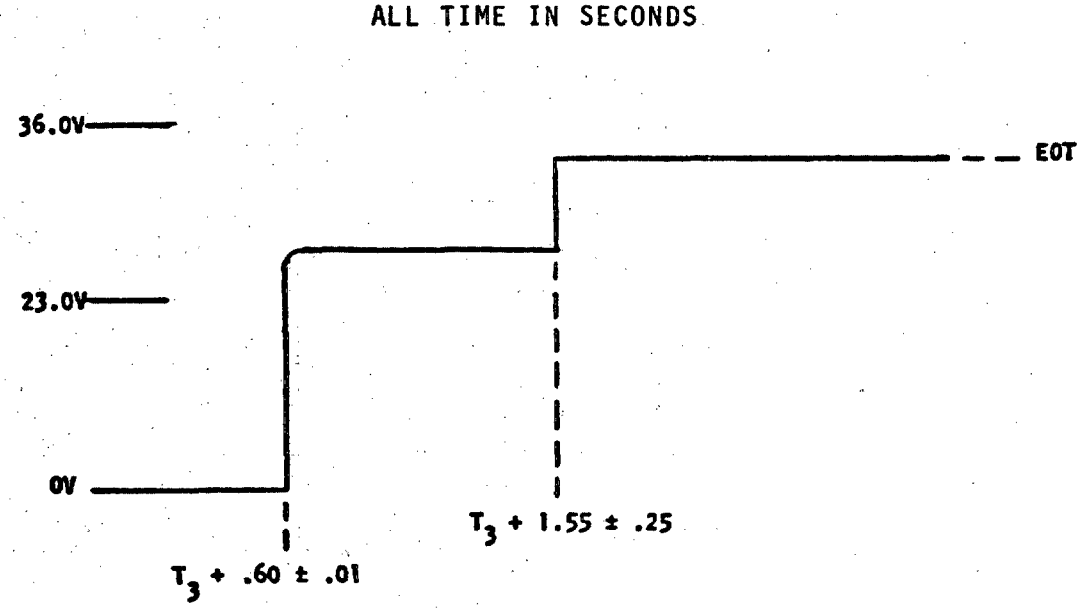
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REMARKS: SIGNAL SHALL BE ZERO AT ALL OTHER TIMES.

Figure 20. Voltage Output 4A and 4B (AK) or Output 4A and 4B (WH)

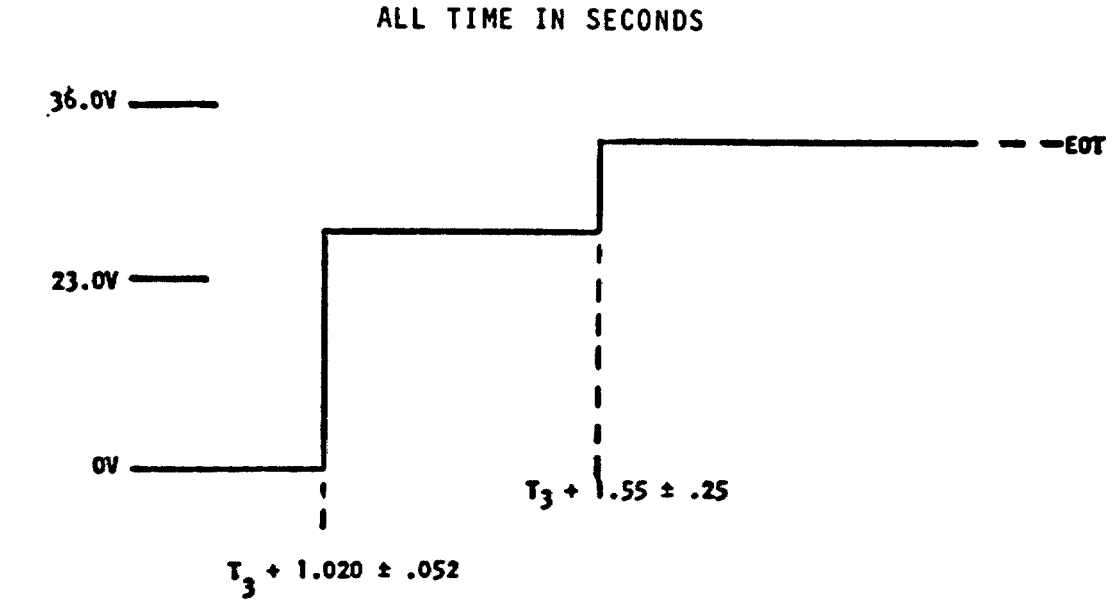
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REMARKS: SIGNAL SHALL BE ZERO AT ALL OTHER TIMES.

Figure 21. Voltage Output 5A and 5B (AK) or Output 5A and 5B (WH)

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REMARKS: SIGNAL SHALL BE ZERO AT ALL OTHER TIMES.

Figure 22. Voltage Output 6A and 6B (AK) or Output 6A and 6B (WH)

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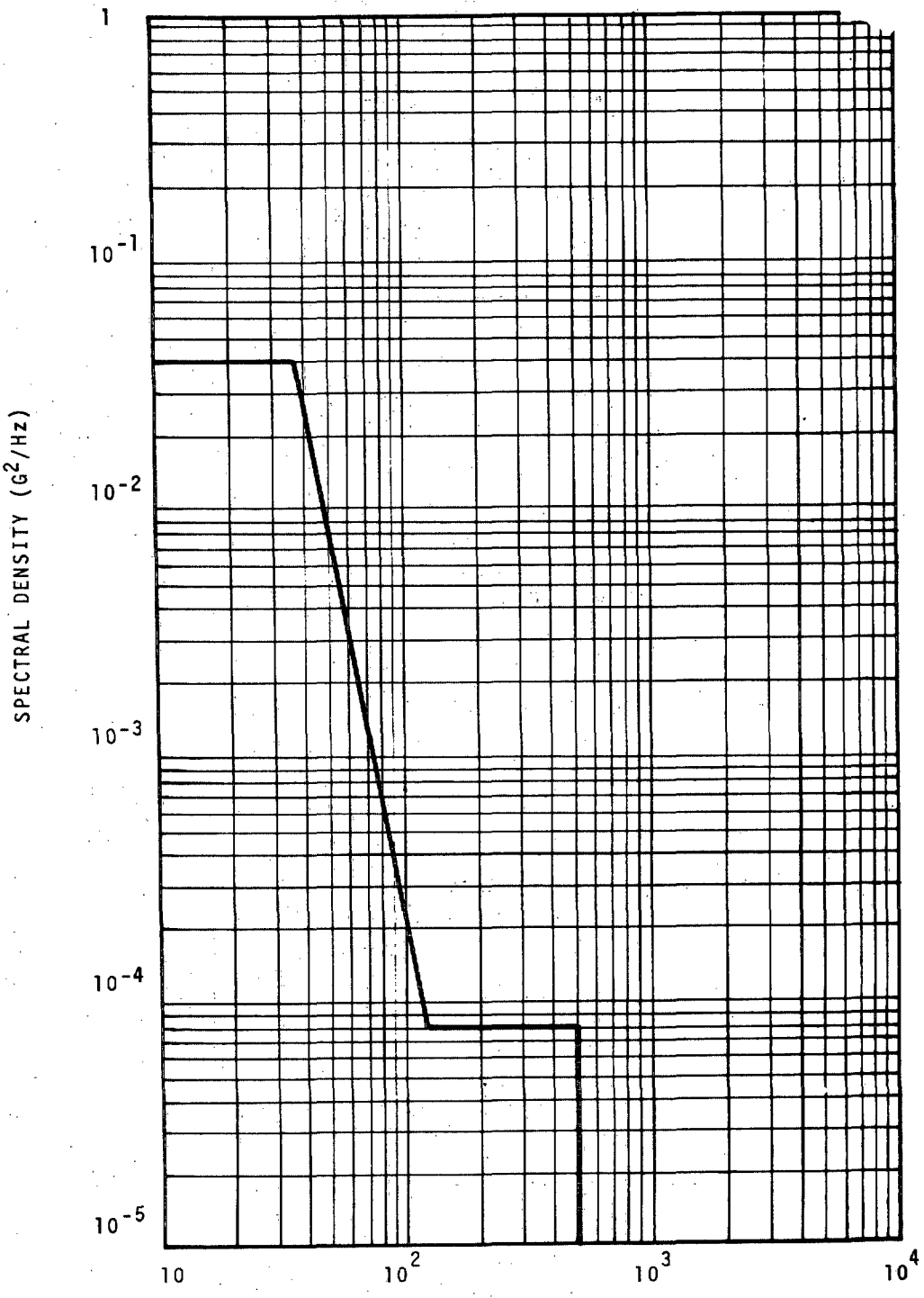


Figure 23. Transportation Random Vibration Y Axis

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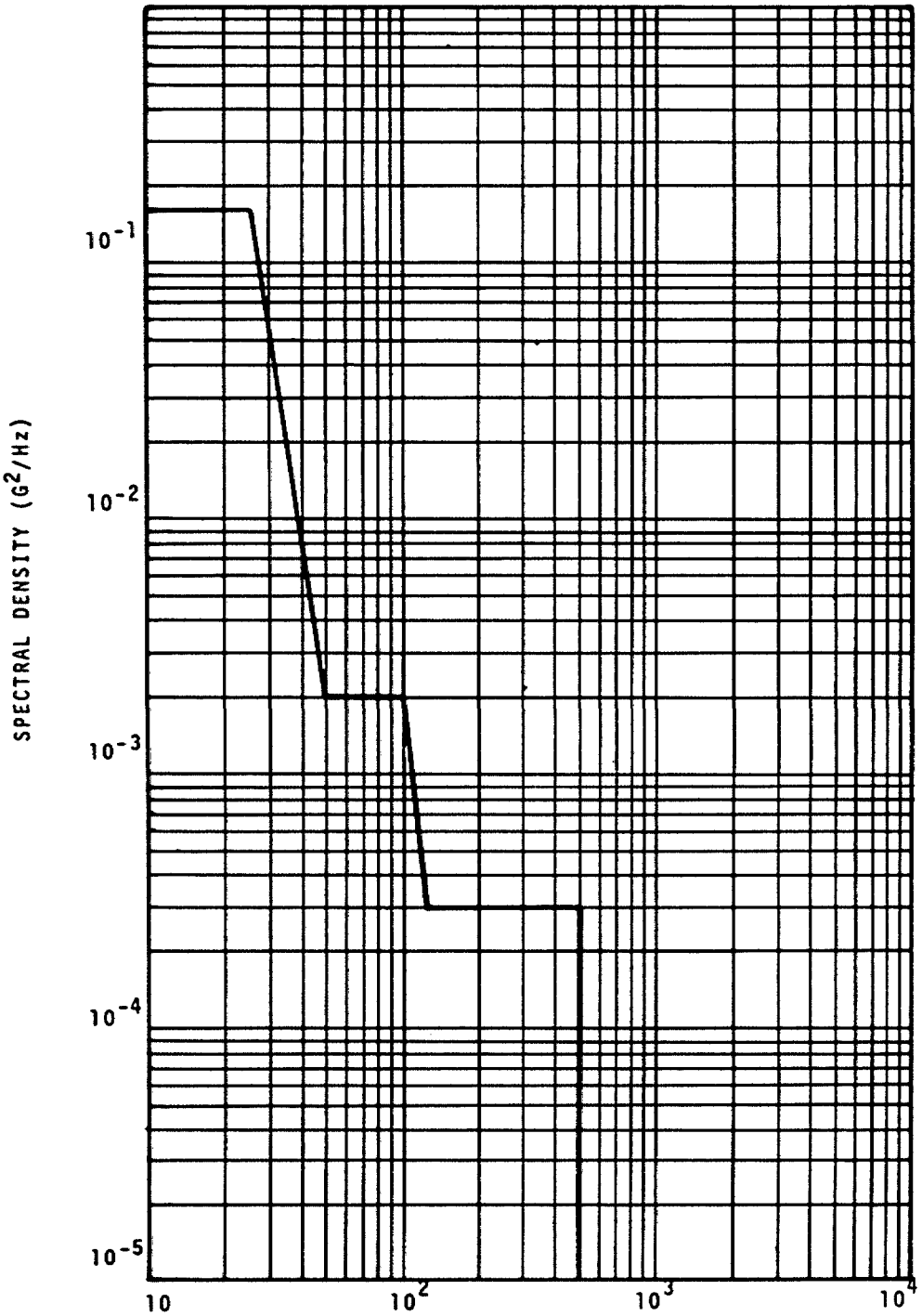


Figure 24. Transportation Random Vibration Z Axis

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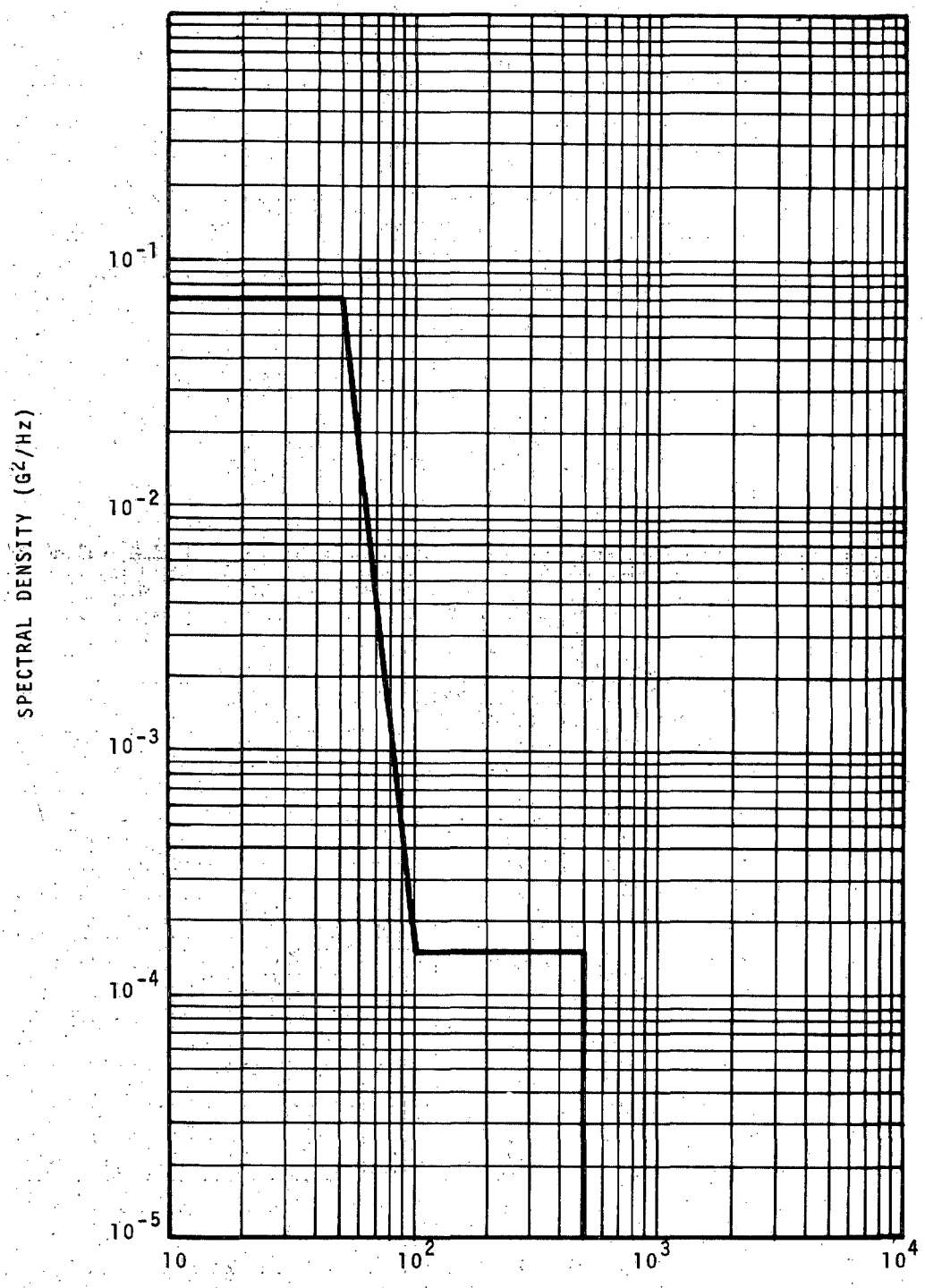


Figure 25. Transportation Random Vibration X Axis

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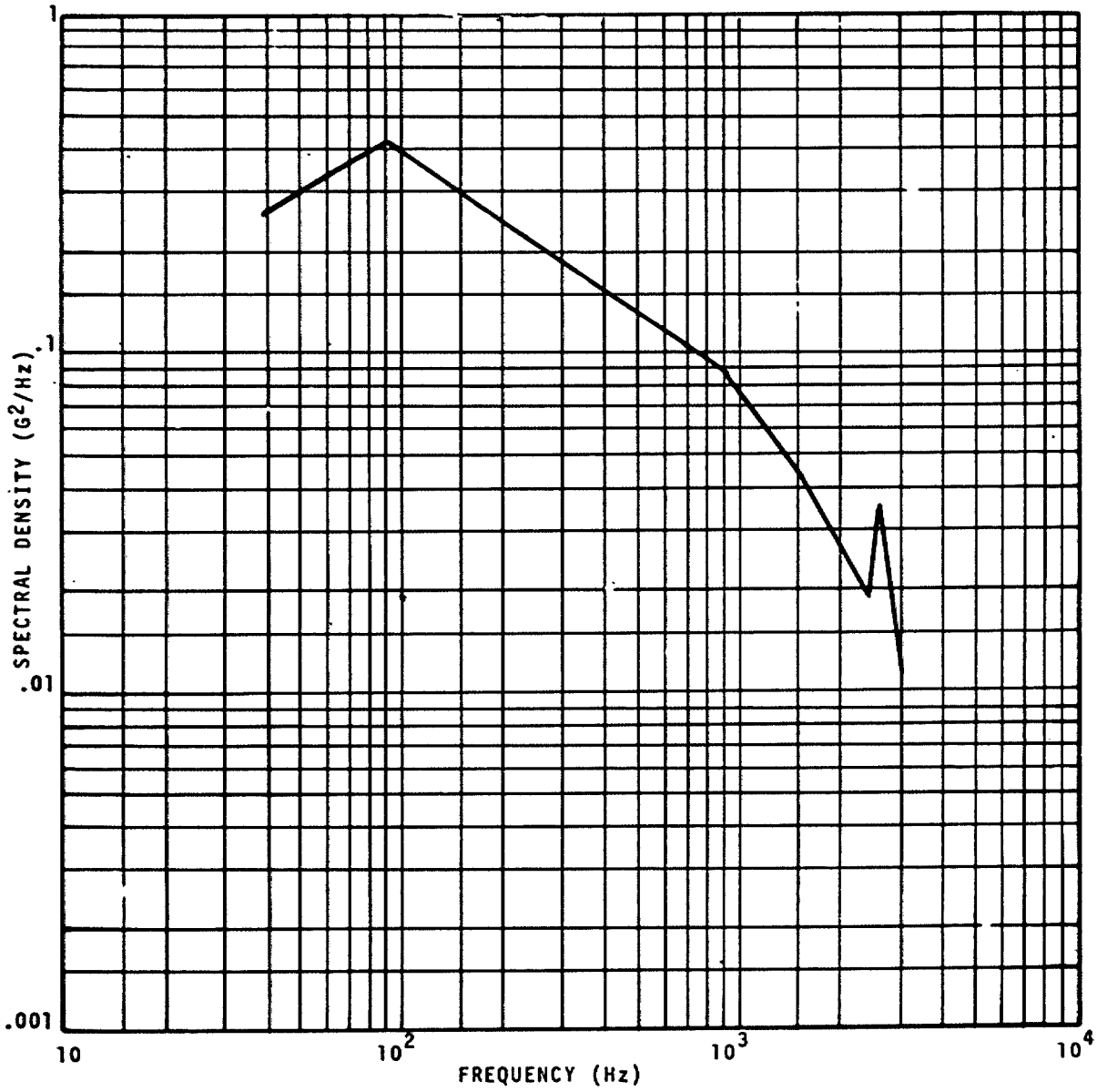


Figure 26. Flight Random Vibration All Axes

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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No. 22-R255

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